

Artificial Intelligence

COSC2129/2723/3118

Week 11: Reinforcement Learning

Slides are adapted from Prof. Sebastian Sardina

RMIT University

Artificial Intelligence – Road map

8 Markov Decision Processes

9 Bayesian Reasoning

10 Reinforcement Learning

11 AI Philosophy + Course recap (week 12)

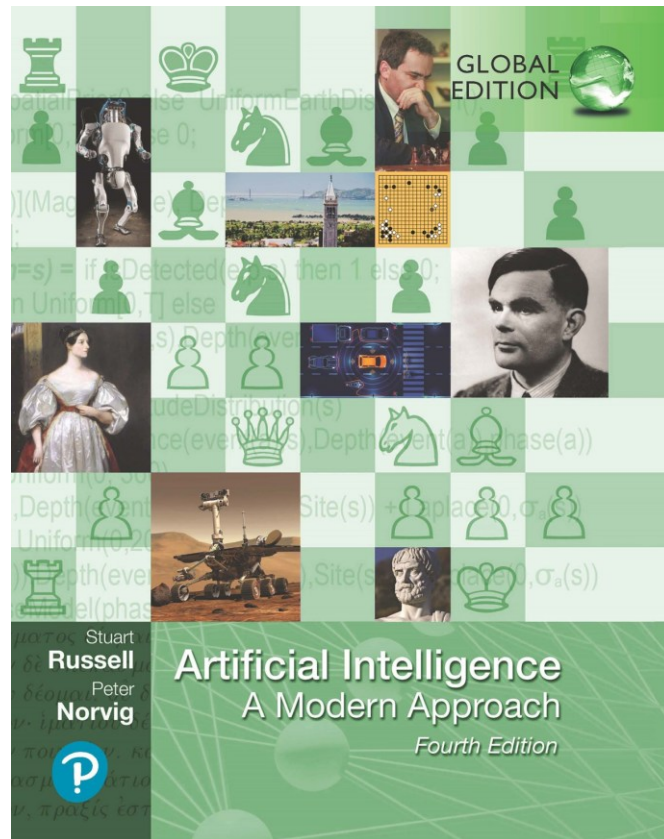


Today

- Assignment 3 has been released

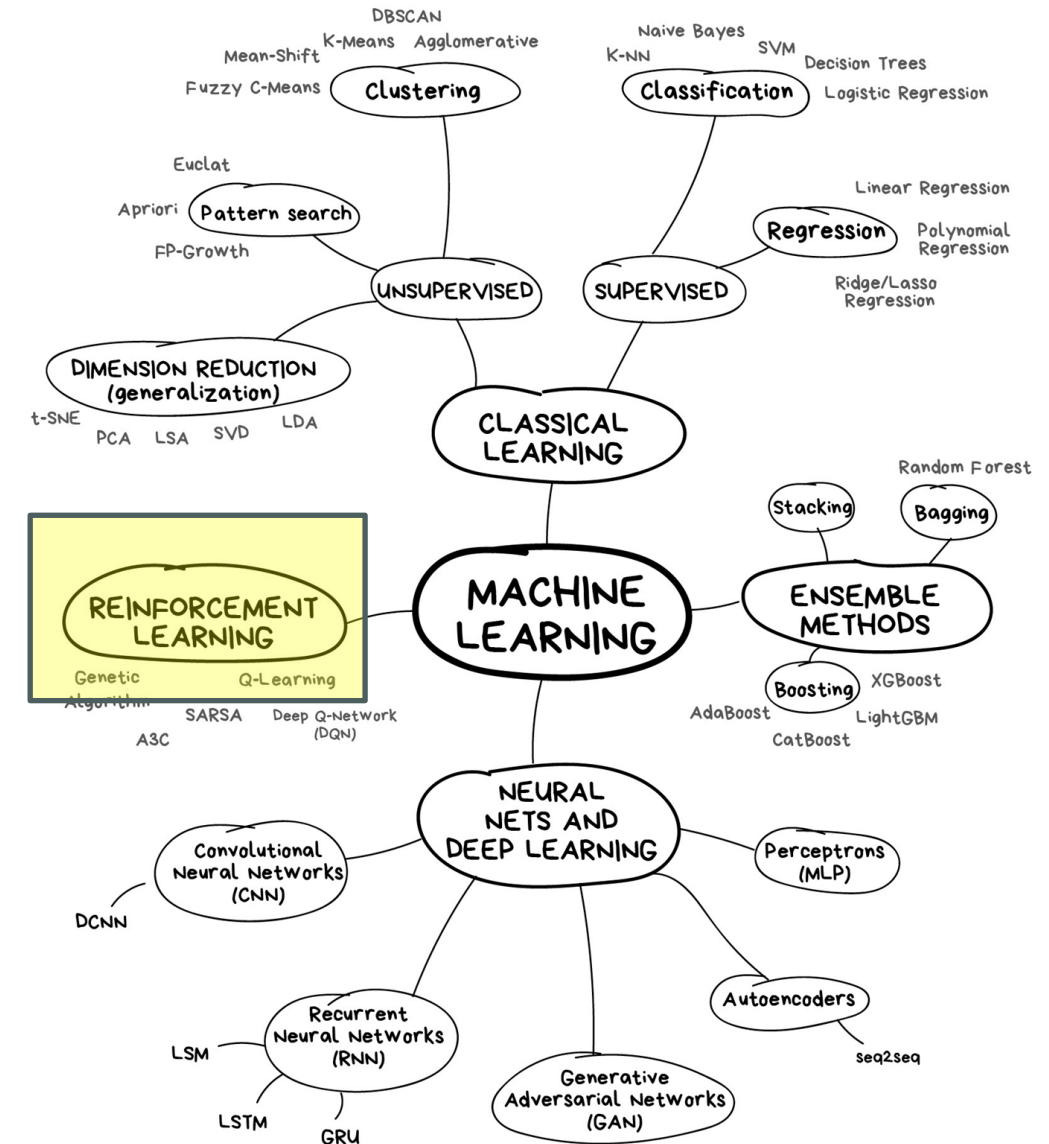
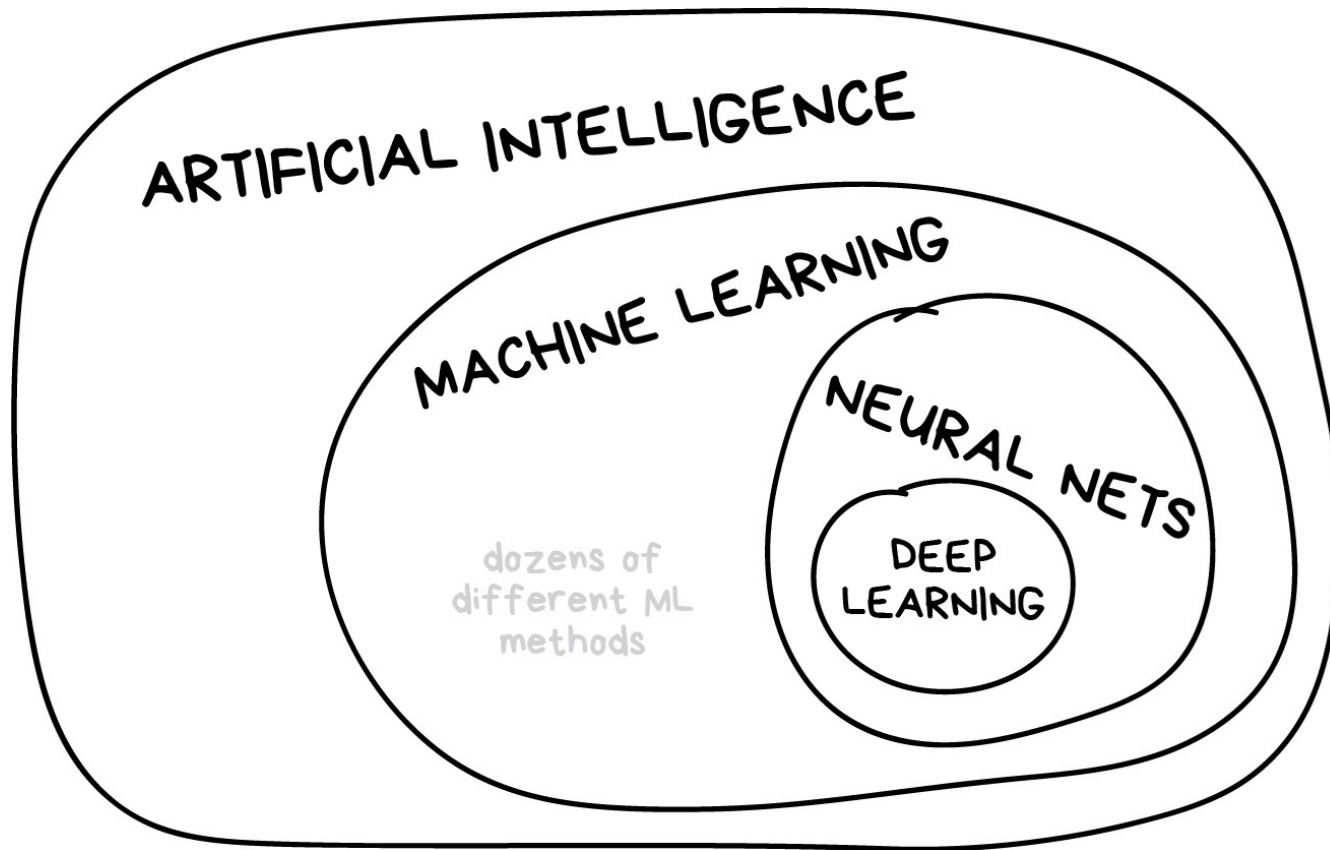
Tasks:

- Implementation
 - Reflection on your implementation
 - Declaration of AI use
-
- Questions?



23 Reinforcement Learning	840
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Where are we in this zoo?



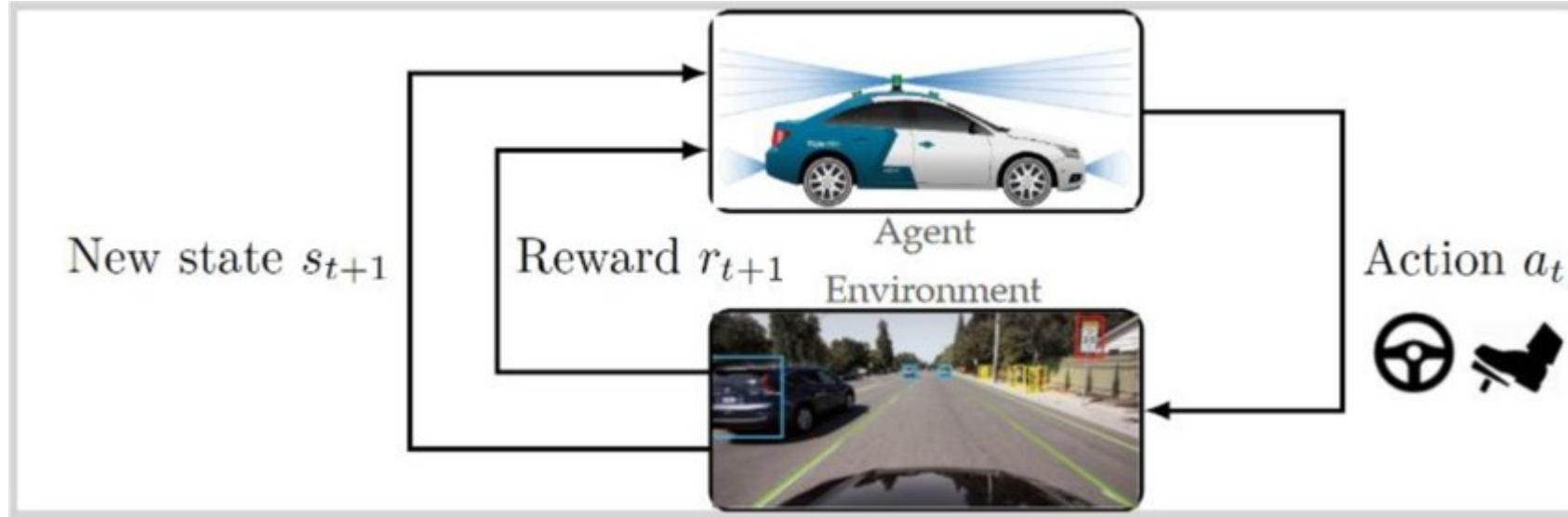
Reinforcement Learning

- From human learning



Reinforcement Learning

- To autonomous driving car



Reinforcement Learning

- Robot



RL + Model-based Control: Using On-demand Optimal Control to Learn Versatile Legged Locomotion

To Alphago



A movie about one of the most famous applications of Deep RL

Should watch [Youtube](#)

CS 188: Artificial Intelligence

Reinforcement Learning



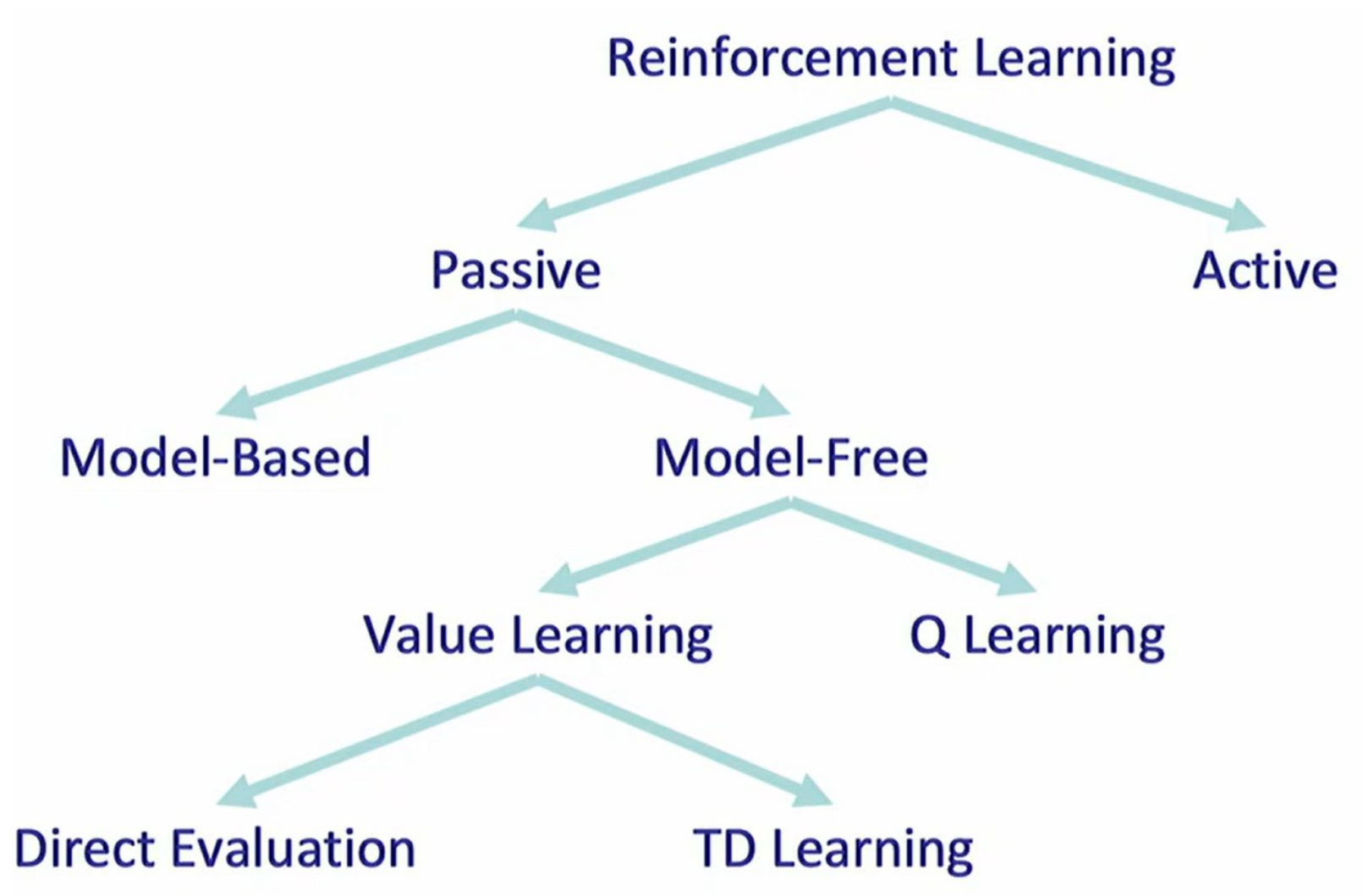
University of California, Berkeley

[These slides were created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. All CS188 materials are available at <http://ai.berkeley.edu>.]

Cool Topics

- Model-based vs Model-free learning
- Direct evaluation and ADP for Model-based learning
- Passive vs Active learning for Model-free learning
- Temporal difference (TD) for model-free passive learning
- Q-learning for active learning: learn $Q(s, a)$ instead of $V(s)$
- TD Q-learning
- Exploration vs Exploitation: random vs exploration functions
- Approximate Q-learning (instead of tabular Q-learning).

Cool Topics

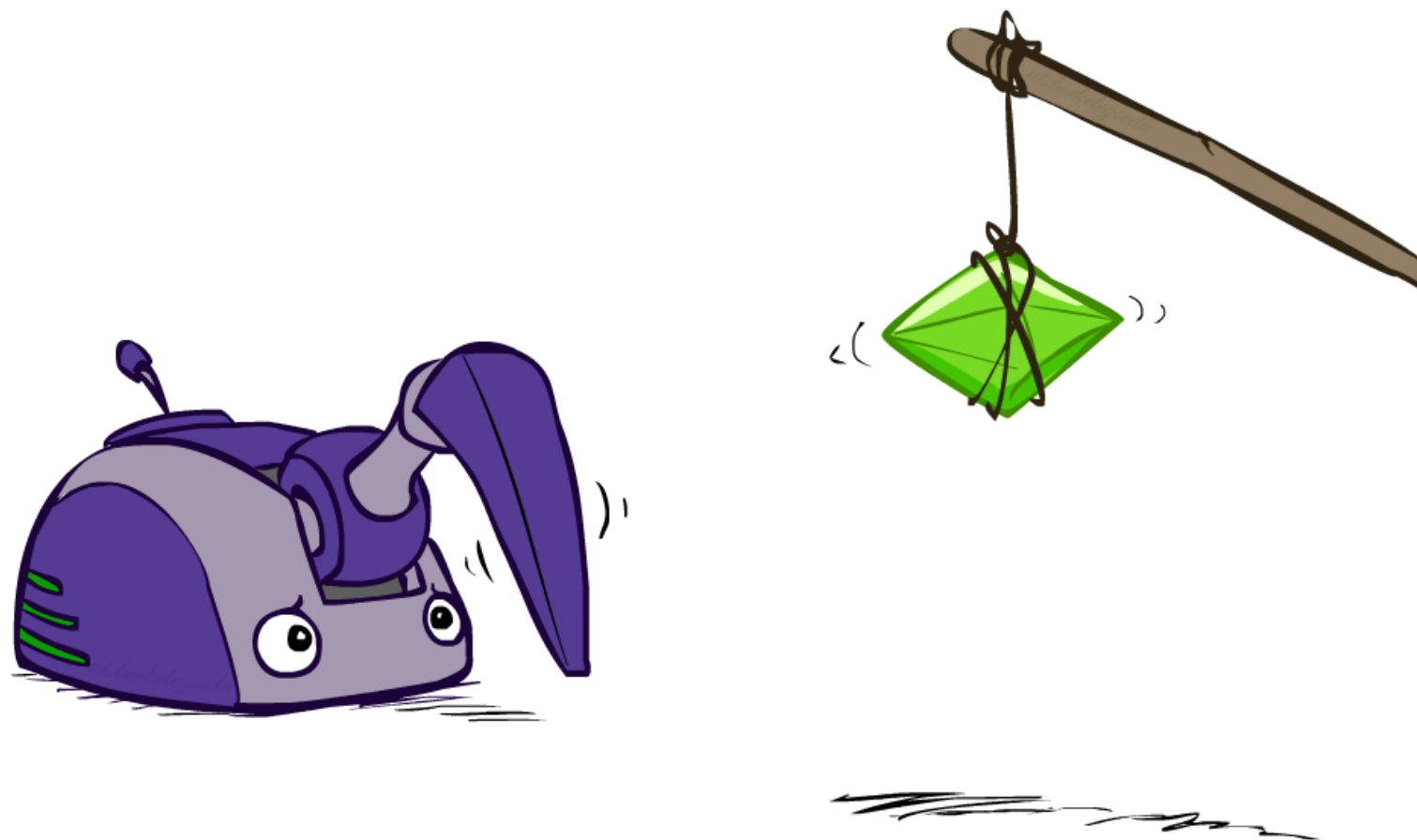


Reinforcement Learning

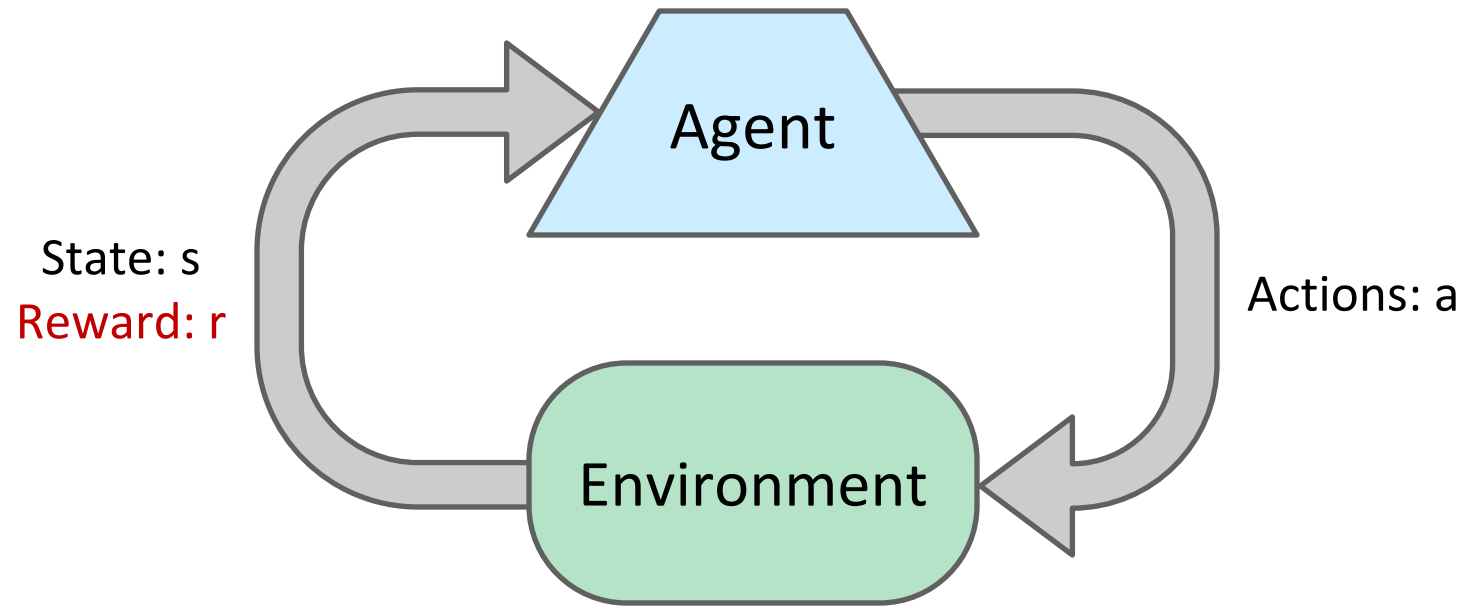
- Video lectures in the slides: did you watch?



Reinforcement Learning



Reinforcement Learning



- **Basic idea:**

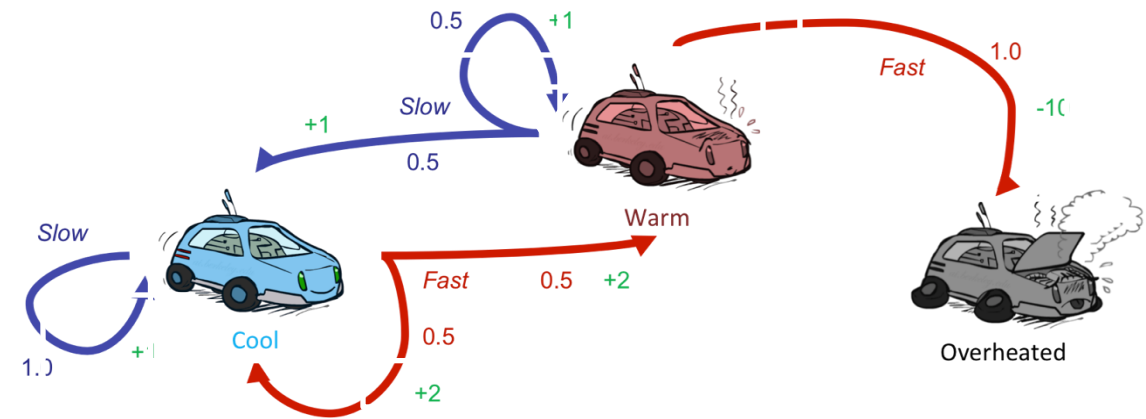
- Receive feedback in the form of **rewards**
- Agent's utility is defined by the reward function
- Must (learn to) act so as to **maximize expected rewards**
- All learning is based on observed samples of outcomes!

Reinforcement Learning

- Still assume a Markov decision process (MDP):

- A set of states $s \in S$
- A set of actions (per state) $a \in A$
- A model $T(s,a,s') = P(s' \mid s, a)$
- A reward function $R(s,a,s')$

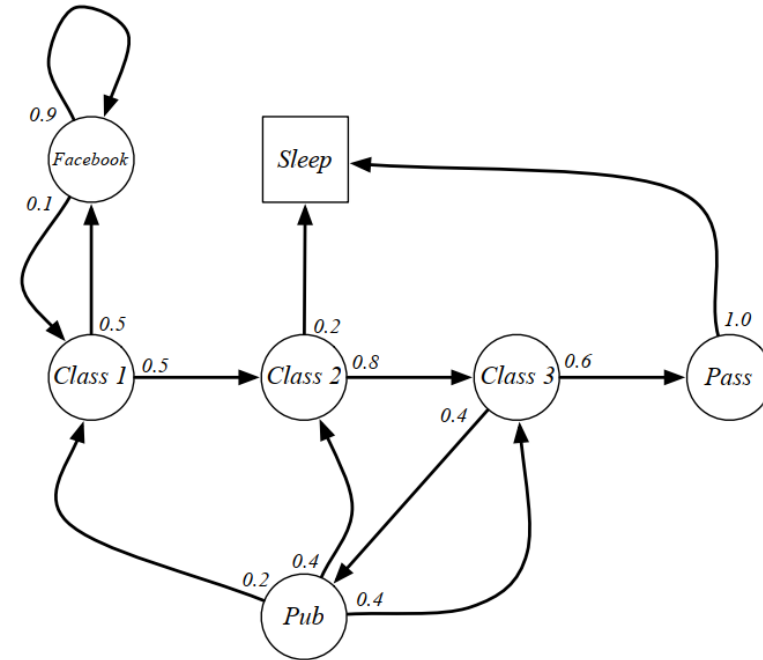
- Still looking for a policy $\pi(s)$



- What is learning in MDPs?

Reinforcement Learning

- Still assume a Markov decision process (MDP):
 - A set of states $s \in S$
 - A set of actions (per state) $a \in A$
 - A model $T(s,a,s') = P(s' \mid s, a)$
 - A reward function $R(s,a,s')$
- Still looking for a policy $\pi(s)$
- New twist: don't know T or R
 - I.e. we don't know which states are good or what the actions do
 - Must actually try actions and states out to learn



Example: Learning to Walk



Initial



A Learning Trial

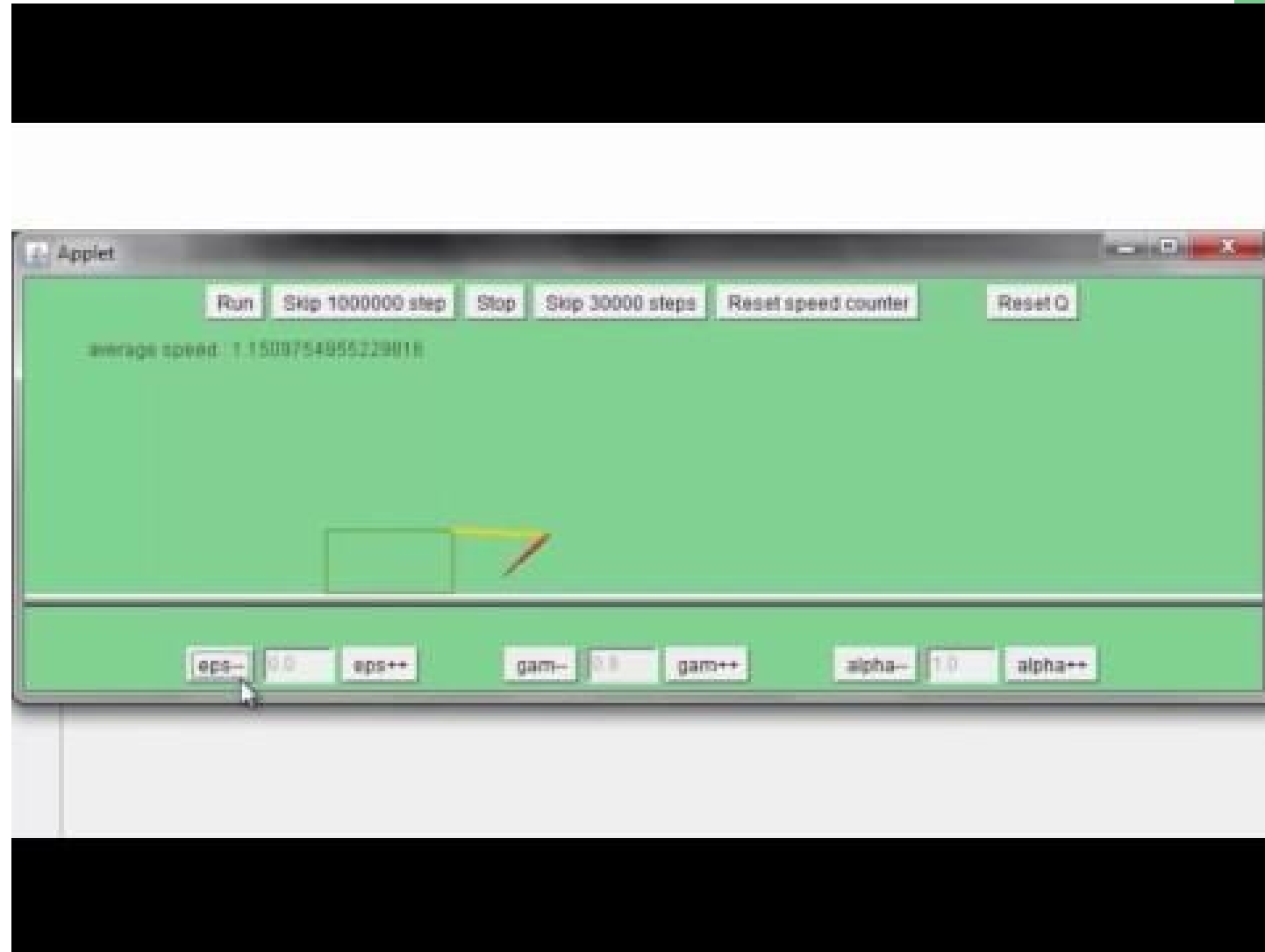
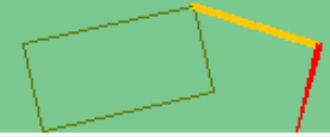


After Learning [1K Trials]

Example: Toddler Robot

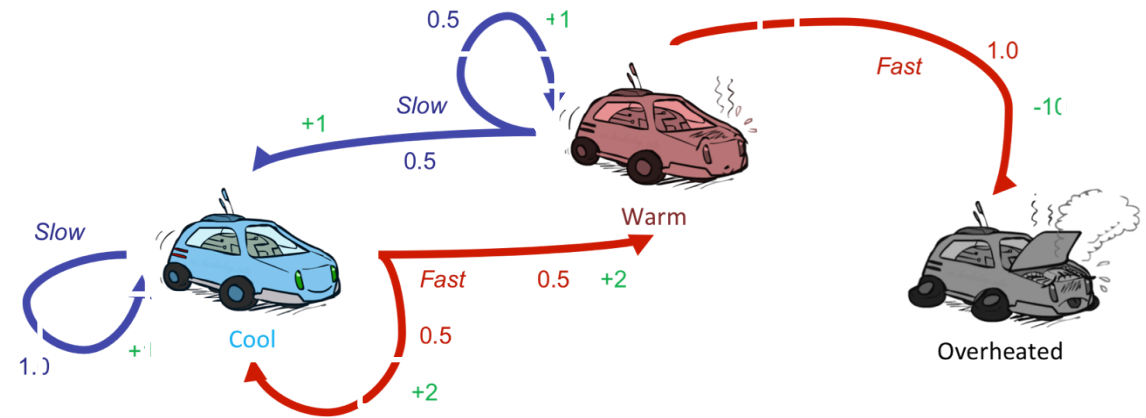


The Crawler!

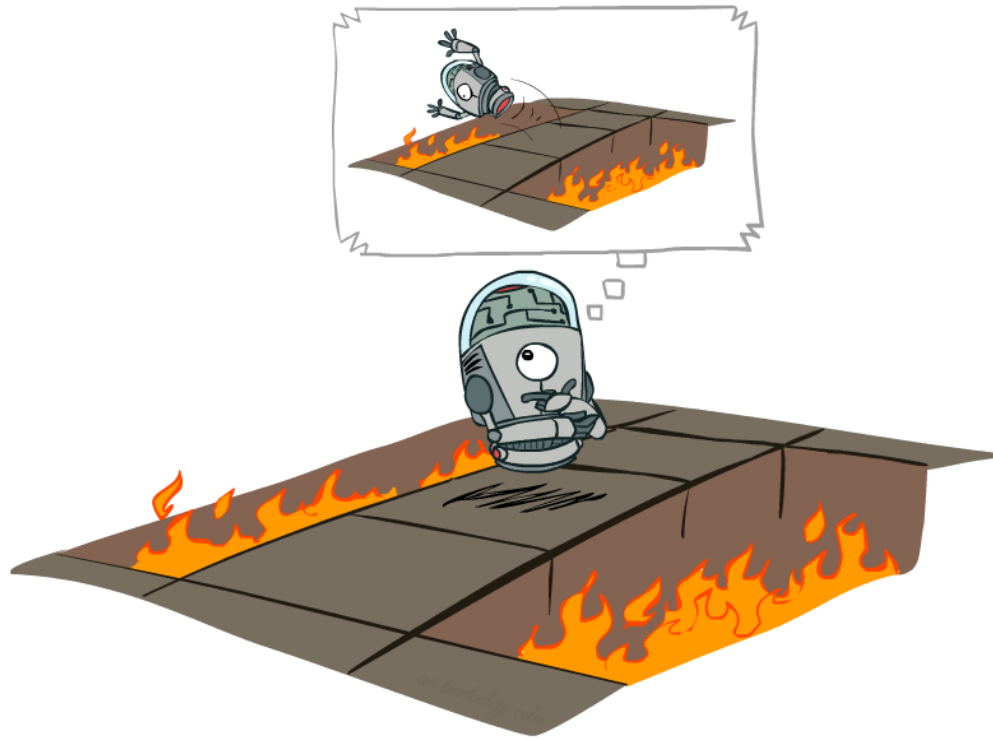


Reinforcement Learning

- Still assume a Markov decision process (MDP):
 - A set of states $s \in S$
 - A set of actions (per state) $a \in A$
 - A model $T(s,a,s') = P(s' \mid s, a)$
 - A reward function $R(s,a,s')$
- Still looking for a policy $\pi(s)$
- New twist: don't know T or R
 - I.e. we don't know which states are good or what the actions do
 - Must actually try actions and states out to learn



Offline (MDPs) vs. Online (RL)

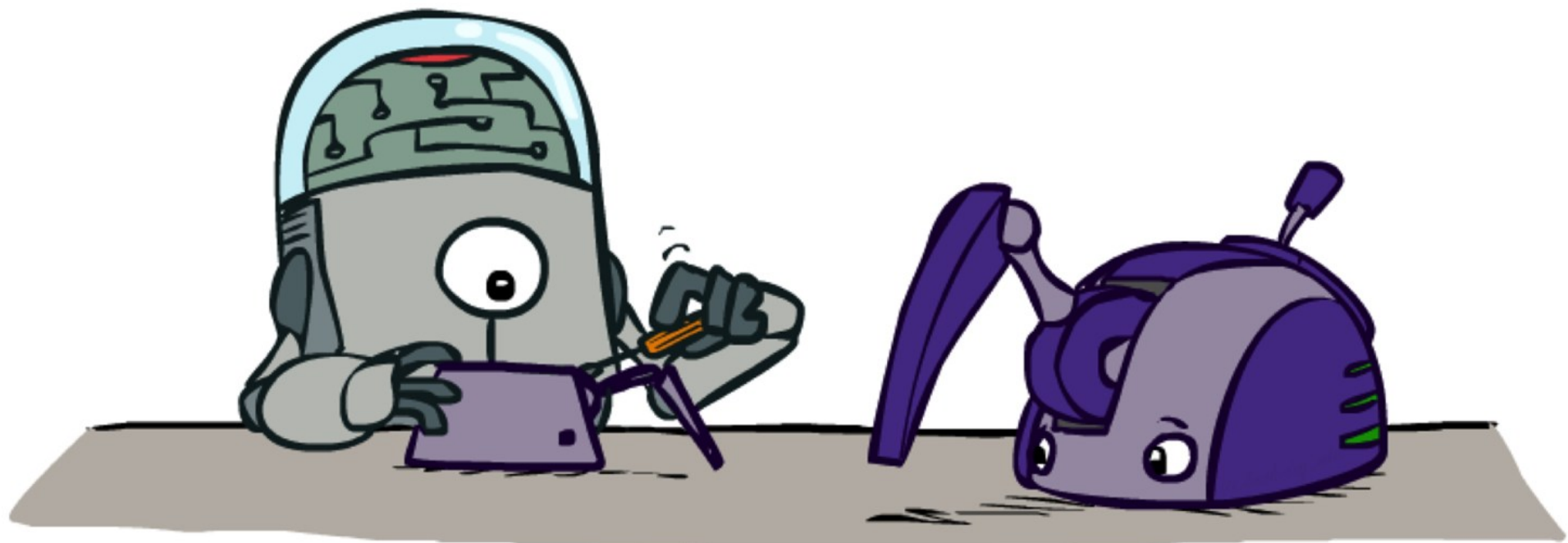


Offline Solution



Online Learning

Model-Based Learning



Model-based vs Model-free Learning: Expected Age

Goal: Compute expected age of class students

$P(a)$: probability of age a in the class/group - ‘the model’

Known $P(A)$

$$E[A] = \sum_a P(a) \cdot a \quad = 0.35 \times 20 + \dots$$

Without $P(A)$, instead collect samples $[a_1, a_2, \dots, a_N]$

Unknown $P(A)$: “Model Based”

$$\hat{P}(a) = \frac{\text{num}(a)}{N}$$

$$E[A] \approx \sum_a \hat{P}(a) \cdot a$$

Why does this work? Because eventually you learn the right model.

Unknown $P(A)$: “Model Free”

$$E[A] \approx \frac{1}{N} \sum_i a_i$$

Why does this work? Because samples appear with the right frequencies.

Adaptive Dynamic Programming: Model-Based Learning

- **Model-Based Idea:**
 - Learn an approximate model based on experiences
 - Solve for values as if the learned model were correct
- **Step 1: Learn empirical MDP model**
 - Count outcomes s' for each s, a
 - Normalize to give an estimate of $\hat{T}(s, a, s')$
 - Discover each $\hat{R}(s, a, s')$ when we experience (s, a, s')
- **Step 2: Solve the learned MDP**
 - For example, use value iteration, as before, to estimate value U



How to estimate $T(s,a,s')$?

$$T(s,a,s') = \frac{\text{number of times we seen } (s,a,s')}{\text{number of times we seen } (s,a)}$$

E.g., if we have outcomes (C, east, D) ,
 (C, east, A) , (C, east, D) , then:

$$T(C, \text{east}, D) = \frac{2}{3}$$

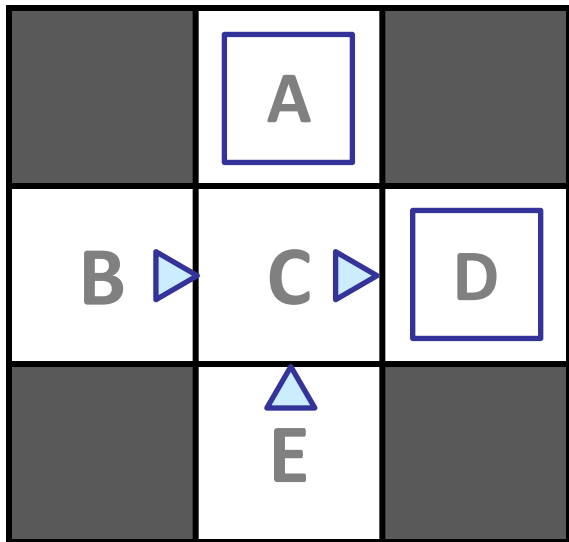
$$T(C, \text{east}, A) = \frac{1}{3}$$

$$T(C, \text{east}, E) = \frac{0}{3}, T(C, \text{east}, B) = \frac{0}{3}$$

	A	
B	C	D
	E	

Example: Model-Based Learning

Input Policy π



Assume: $\gamma = 1$

Observed Episodes (Training)

Episode 1

B, east, C, -1
C, east, D, -1
D, exit, x, +10

Episode 2

B, east, C, -1
C, east, D, -1
D, exit, x, +10

Episode 3

E, north, C, -1
C, east, D, -1
D, exit, x, +10

Episode 4

E, north, C, -1
C, east, A, -1
A, exit, x, -10

Learned Model

$$\hat{T}(s, a, s')$$

T(B, east, C) = 1.00
T(C, east, D) = 0.75
T(C, east, A) = 0.25
...

$$\hat{R}(s, a, s')$$

R(B, east, C) = -1
R(C, east, D) = -1
R(D, exit, x) = +10
...

Adaptive Dynamic Programming

function *Passive-ADP-Agent*(s', r')

Inputs: sample (current state s' , reward signal r')

Global: mdp (model T , rewards R), table of utilities U , table of frequencies for state-action pairs N_{sa} , table of frequencies for state-action-state triples $N_{sas'}$, the previous state, action (s, a)

if s' is new, then $U[s'] = r'; R[s'] = r'$

if s is not null, then

increment $N_{sa}[s, a]$ and $N_{sas'}[s, a, s']$

for each t such that $N_{sas'}[s, a, t] > 0$

$T[s, a, t] = N_{sas'}[s, a, t] / N_{sa}[s, a]$

$U = \text{policy-evaluation}(\pi, U, \text{mdp})$

Adaptive Dynamic Programming in ALMA 4G

```

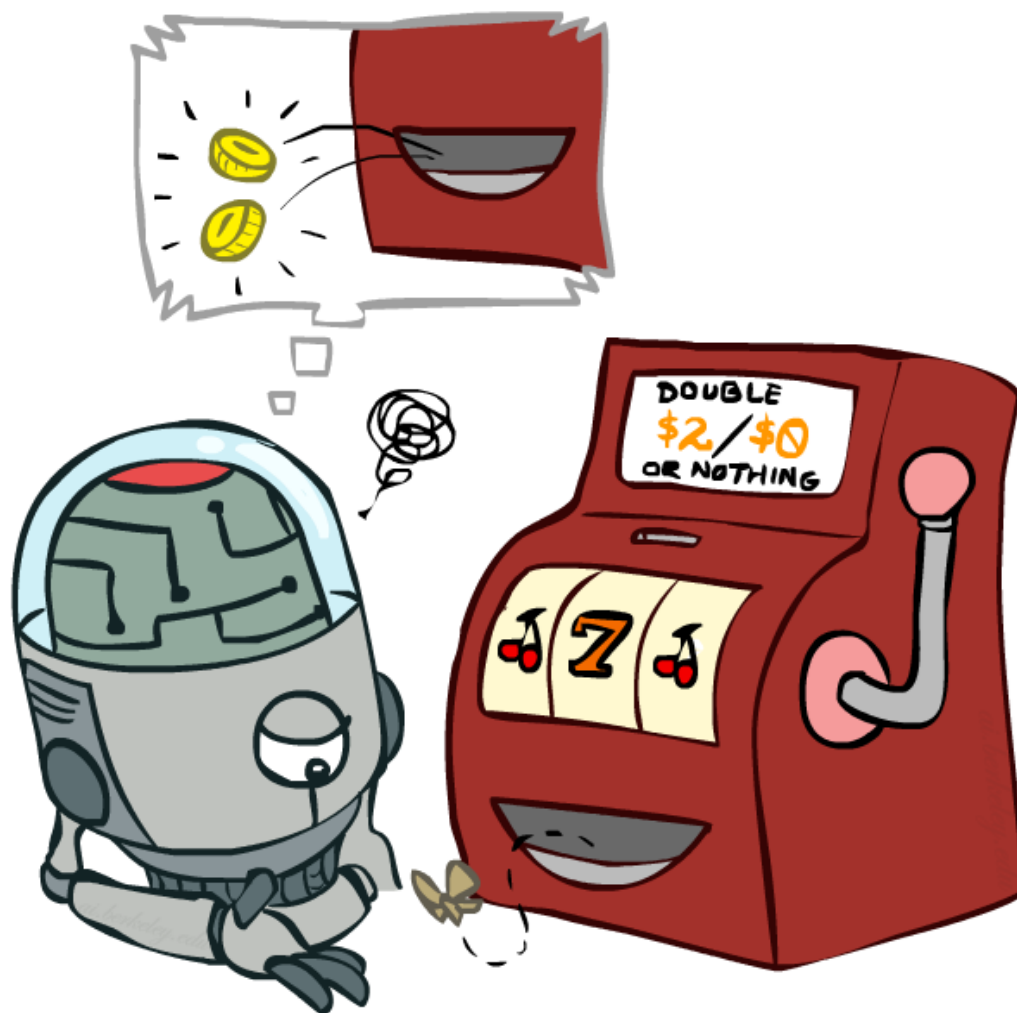
function PASSIVE-ADP-LEARNER(percept) returns an action
  inputs: percept, a percept indicating the current state  $s'$  and reward signal  $r$ 
  persistent:  $\pi$ , a fixed policy
                $mdp$ , an MDP with model  $P$ , rewards  $R$ , actions  $A$ , discount  $\gamma$ 
                $U$ , a table of utilities for states, initially empty
                $N_{s'|s,a}$ , a table of outcome count vectors indexed by state and action, initially zero
                $s, a$ , the previous state and action, initially null

  if  $s'$  is new then  $U[s'] \leftarrow 0$ 
  if  $s$  is not null then
    increment  $N_{s'|s,a}[s, a][s']$ 
     $R[s, a, s'] \leftarrow r$ 
    add  $a$  to  $A[s]$ 
     $\mathbf{P}(\cdot \mid s, a) \leftarrow \text{NORMALIZE}(N_{s'|s,a}[s, a])$ 
     $U \leftarrow \text{POLICYEVALUATION}(\pi, U, mdp)$ 
     $s, a \leftarrow s', \pi[s']$ 
  return  $a$ 

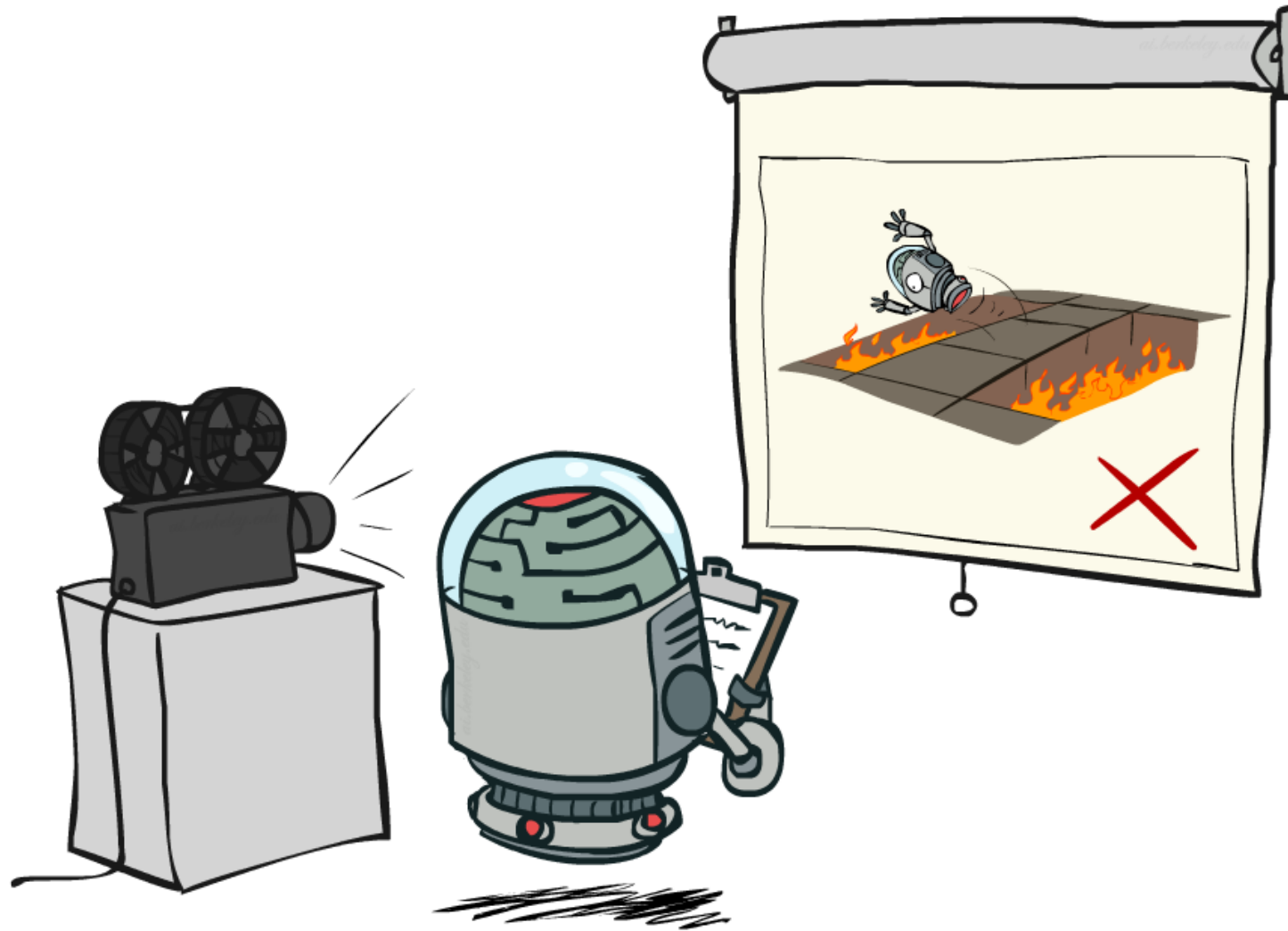
```

Figure 23.2 A passive reinforcement learning agent based on adaptive dynamic programming. The agent chooses a value for γ and then incrementally computes the P and R values of the MDP. The POLICY-EVALUATION function solves the fixed-policy Bellman equations, as described on page 567.

Model-Free Learning



Passive Reinforcement Learning



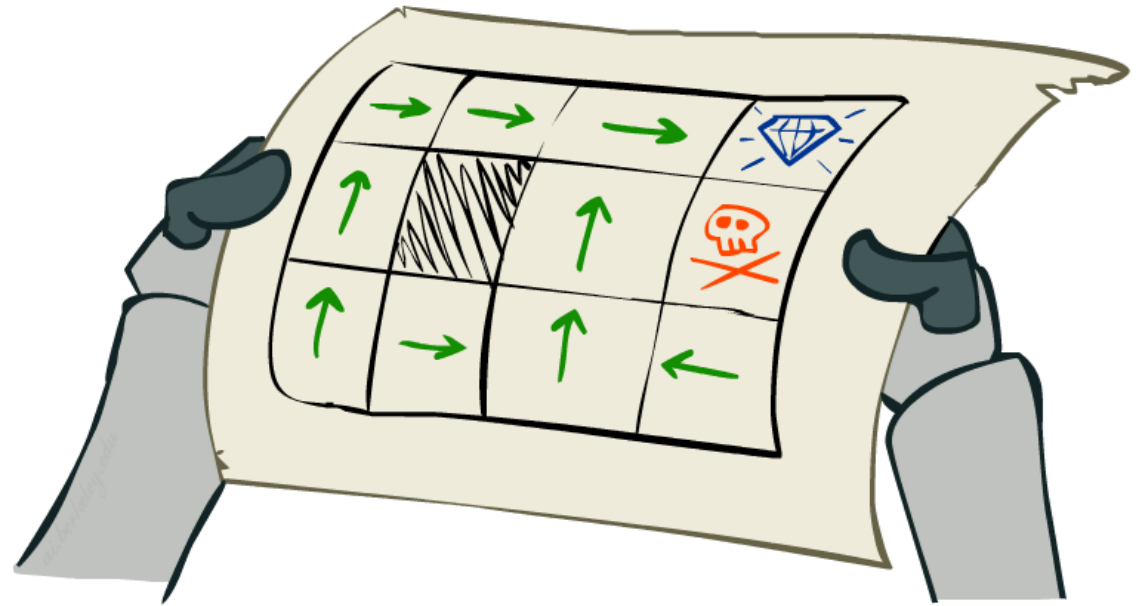
Passive Reinforcement Learning

- Simplified task: policy evaluation

- Input: a fixed policy $\pi(s)$
- You don't know the transitions $T(s,a,s')$
- You don't know the rewards $R(s,a,s')$
- **Goal: learn the state values**

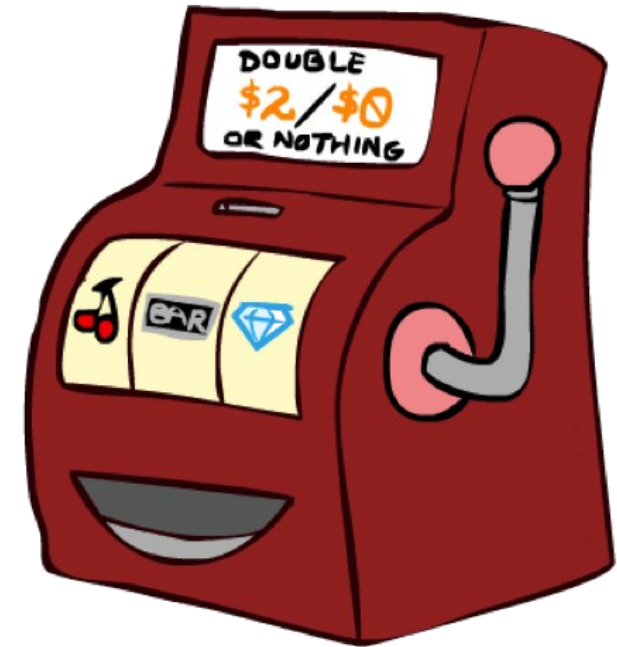
- In this case:

- Learner is “along for the ride”
- No choice about what actions to take
- Just execute the policy and learn from experience
- This is NOT offline planning! You actually take actions in the world.



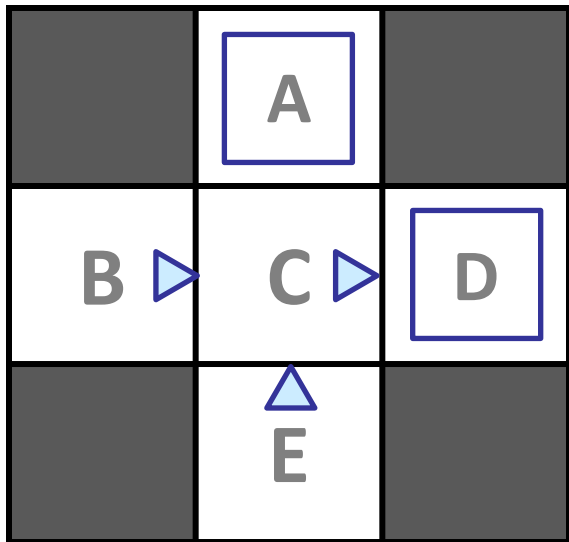
Direct Evaluation

- Goal: Compute values for each state under π
- Idea: Average together observed sample values
 - Act according to π .
 - Every time you visit a state, write down what the sum of discounted rewards turned out to be.
 - Average those samples.
- This is called direct evaluation



Example: Direct Evaluation

Input Policy π



Assume: $\gamma = 1$

Observed Episodes (Training)

Episode 1

B, east, C, -1
C, east, D, -1
D, exit, x, +10

Episode 2

B, east, C, -1
C, east, D, -1
D, exit, x, +10

Episode 3

E, north, C, -1
C, east, D, -1
D, exit, x, +10

Episode 4

E, north, C, -1
C, east, A, -1
A, exit, x, -10

Output Values

	-10 A	
+8 B	+4 C	+10 D
	-2 E	

Problems with Direct Evaluation

- What's good about direct evaluation?
 - It's easy to understand
 - It doesn't require any knowledge of T , R
 - It eventually computes the correct average values, using just sample transitions
- What bad about it?
 - It wastes information about state connections
 - Each state must be learned separately
 - So, it takes a long time to learn

Output Values

	-10 A	
+8 B	+4 C	+10 D
	-2 E	

If B and E both go to C under this policy, how can their values be different?

Why Not Use Policy Evaluation?

- Simplified Bellman updates calculate V for a fixed policy:

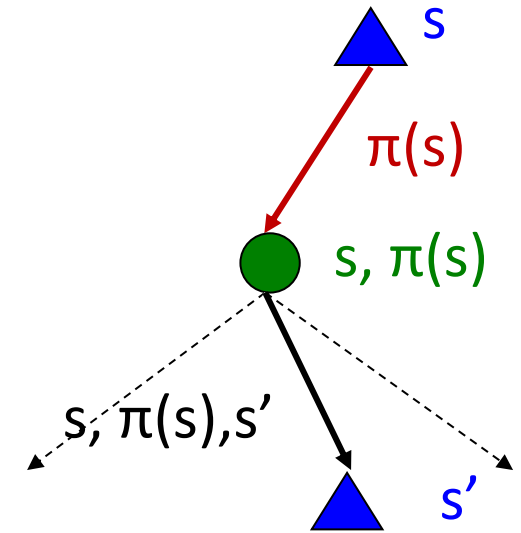
- Each round, replace V with a one-step-look-ahead layer over V

$$V_0^\pi(s) = 0$$

$$V_{k+1}^\pi(s) \leftarrow \sum_{s'} T(s, \pi(s), s') [R(s, \pi(s), s') + \gamma V_k^\pi(s')]$$

- This approach fully exploited the connections between the states
- Unfortunately, we need T and R to do it!

- Key question: how can we do this update to V without knowing T and R ?
 - In other words, how do we take a weighted average without knowing the weights?



Sample-Based Policy Evaluation?

- We want to improve our estimate of V by computing these averages:

$$V_{k+1}^{\pi}(s) \leftarrow \sum_{s'} T(s, \pi(s), s') [R(s, \pi(s), s') + \gamma V_k^{\pi}(s')]$$

- Idea: Take samples of outcomes s' (by doing the action!) and average

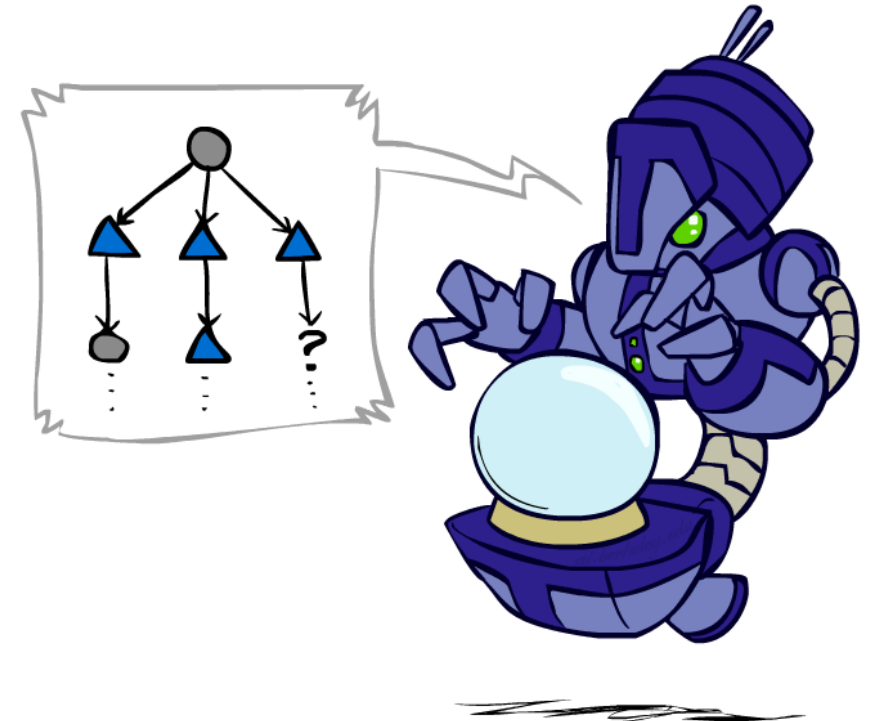
$$sample_1 = R(s, \pi(s), s'_1) + \gamma V_k^{\pi}(s'_1)$$

$$sample_2 = R(s, \pi(s), s'_2) + \gamma V_k^{\pi}(s'_2)$$

...

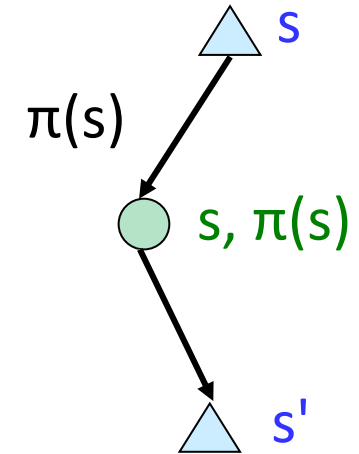
$$sample_n = R(s, \pi(s), s'_n) + \gamma V_k^{\pi}(s'_n)$$

$$V_{k+1}^{\pi}(s) \leftarrow \frac{1}{n} \sum_i sample_i$$



Temporal Difference Learning

- Big idea: learn from every experience!
 - Update $V(s)$ each time we experience a transition (s, a, s', r)
 - Likely outcomes s' will contribute updates more often
- Temporal difference learning of values
 - Policy still fixed, still doing evaluation!
 - Move values toward value of whatever successor occurs: running average



Sample of $V(s)$: $sample = R(s, \pi(s), s') + \gamma V^\pi(s')$

Update to $V(s)$: $V^\pi(s) \leftarrow (1 - \alpha)V^\pi(s) + (\alpha)sample$

Same update: $V^\pi(s) \leftarrow V^\pi(s) + \alpha(sample - V^\pi(s))$

Exponential Moving Average

- Exponential moving average

- The running interpolation update: $\bar{x}_n = (1 - \alpha) \cdot \bar{x}_{n-1} + \alpha \cdot x_n$
- Makes recent samples more important:

$$\bar{x}_n = \frac{x_n + (1 - \alpha) \cdot x_{n-1} + (1 - \alpha)^2 \cdot x_{n-2} + \dots}{1 + (1 - \alpha) + (1 - \alpha)^2 + \dots}$$

- Forgets about the past (distant past values were wrong anyway)
- Decreasing learning rate (alpha) can give converging averages

Example: Temporal Difference Learning

States

	A	
B	C	D
	E	

Assume: $\gamma = 1$, $\alpha = 1/2$

Observed Transitions

B, east, C, -2

	0	
0	0	8
	0	

C, east, D, -2

	0	
-1	0	8
	0	

	0	
-1	3	8
	0	

$$V^\pi(s) \leftarrow (1 - \alpha)V^\pi(s) + \alpha [R(s, \pi(s), s') + \gamma V^\pi(s')]$$

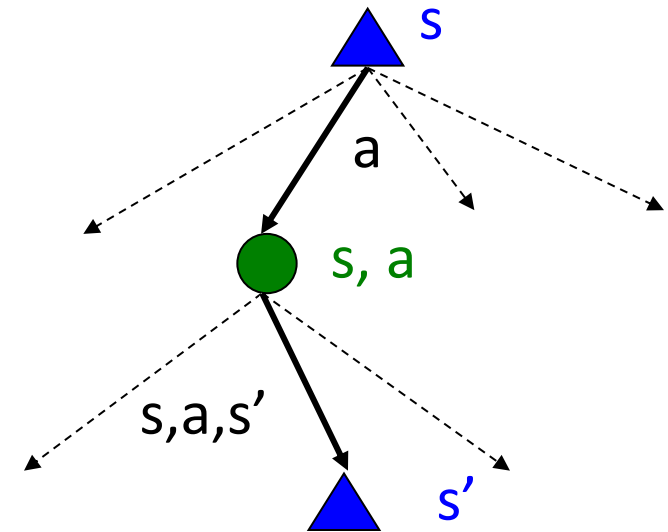
Problems with TD Value Learning

- TD value learning is a model-free way to do policy evaluation, mimicking Bellman updates with running sample averages
- However, if we want to turn values into a (new) policy, we're sunk:

$$\pi(s) = \arg \max_a Q(s, a)$$

$$Q(s, a) = \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V(s')]$$

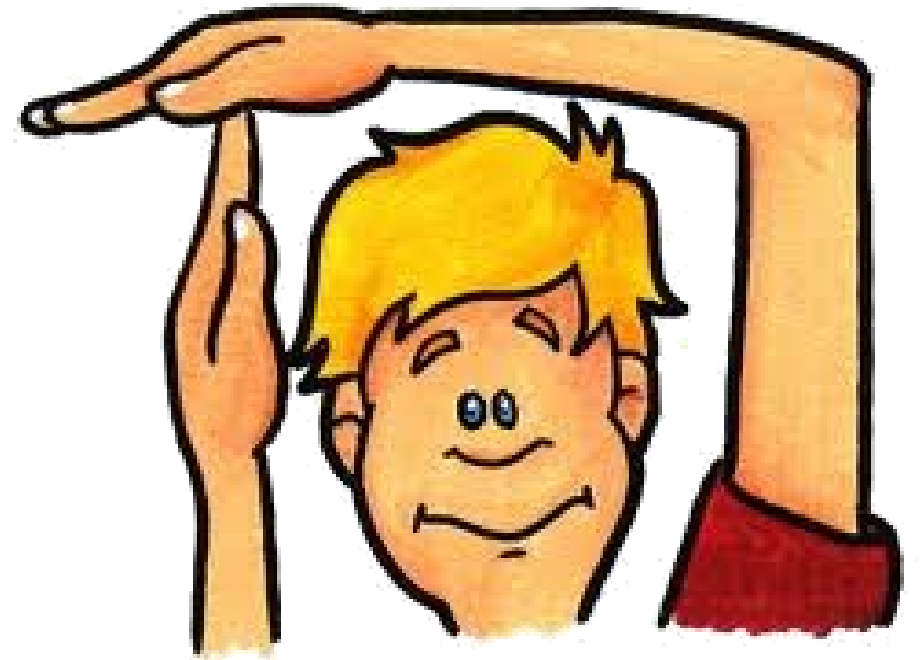
- Idea: learn Q-values, not values
- Makes action selection model-free too!



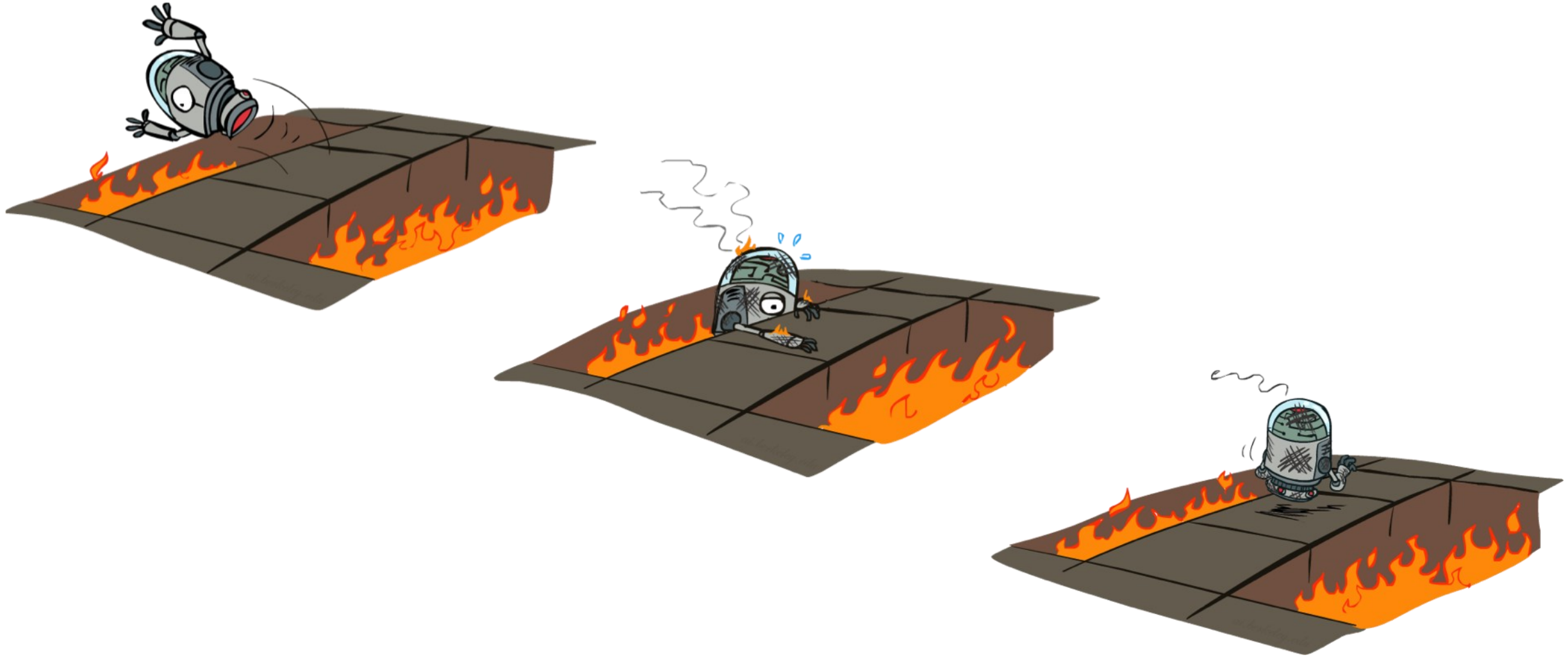
Let's take a break...



Treaty- Yothu Yindi (live at The Big Day Out)

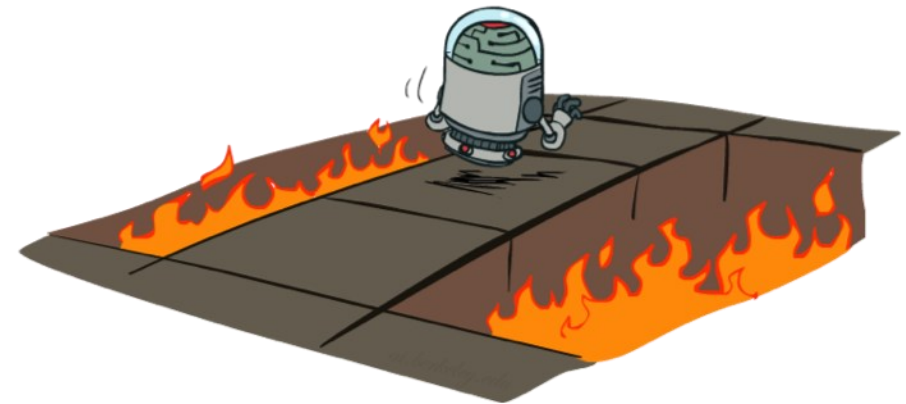


Active Reinforcement Learning



Active Reinforcement Learning

- Full reinforcement learning: optimal policies (like value iteration)
 - You don't know the transitions $T(s,a,s')$
 - You don't know the rewards $R(s,a,s')$
 - You choose the actions now
 - Goal: learn the optimal policy / values
- In this case:
 - Learner makes choices!
 - Fundamental tradeoff: exploration vs. exploitation
 - This is NOT offline planning! You actually take actions in the world and find out what happens...



Detour: Q-Value Iteration

- Value iteration: find successive (depth-limited) values

- Start with $V_0(s) = 0$, which we know is right
- Given V_k , calculate the depth $k+1$ values for all states:

$$V_{k+1}(s) \leftarrow \max_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V_k(s')]$$

- But Q-values are more useful, so compute them instead

- Start with $Q_0(s,a) = 0$, which we know is right
- Given Q_k , calculate the depth $k+1$ q-values for all q-states:

$$Q_{k+1}(s, a) \leftarrow \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma \max_{a'} Q_k(s', a')]$$

Q-Learning

- Q-Learning: sample-based Q-value iteration

$$Q_{k+1}(s, a) \leftarrow \sum_{s'} T(s, a, s') \left[R(s, a, s') + \gamma \max_{a'} Q_k(s', a') \right]$$

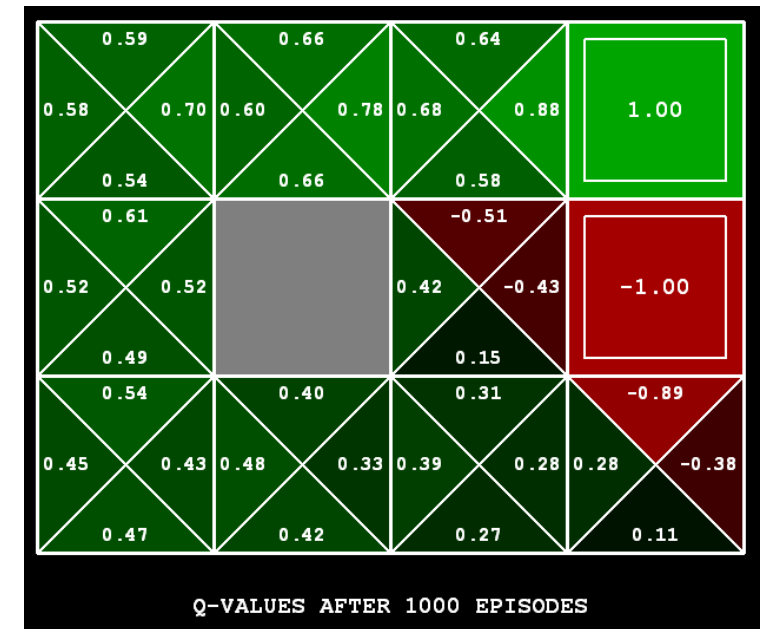
- Learn $Q(s,a)$ values as you go

- Receive a sample (s,a,s',r)
- Consider your old estimate: $Q(s, a)$
- Consider your new sample estimate:

$$sample = R(s, a, s') + \gamma \max_{a'} Q(s', a')$$

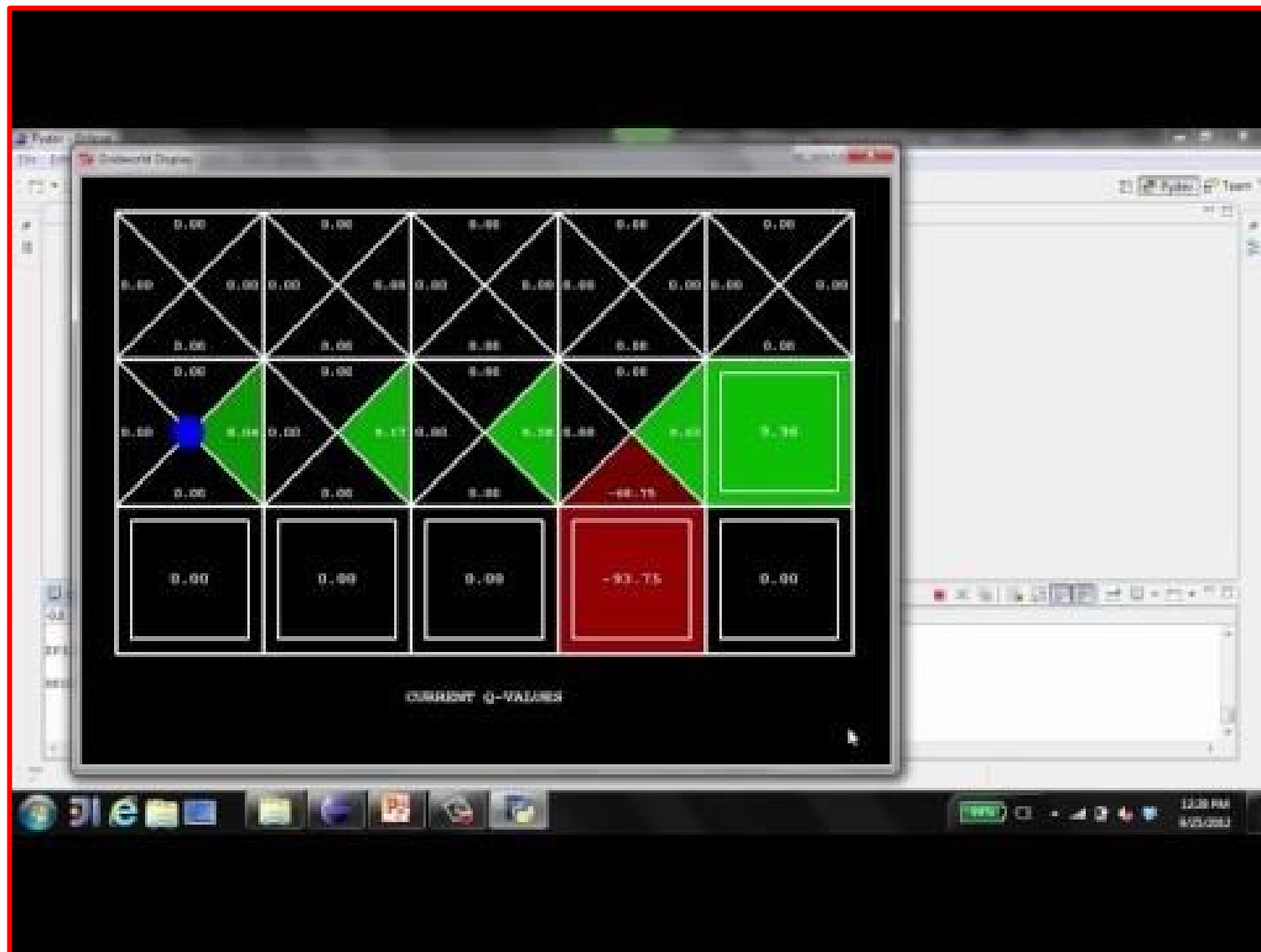
- Incorporate the new estimate into a running average:

$$Q(s, a) \leftarrow (1 - \alpha)Q(s, a) + (\alpha) [sample]$$

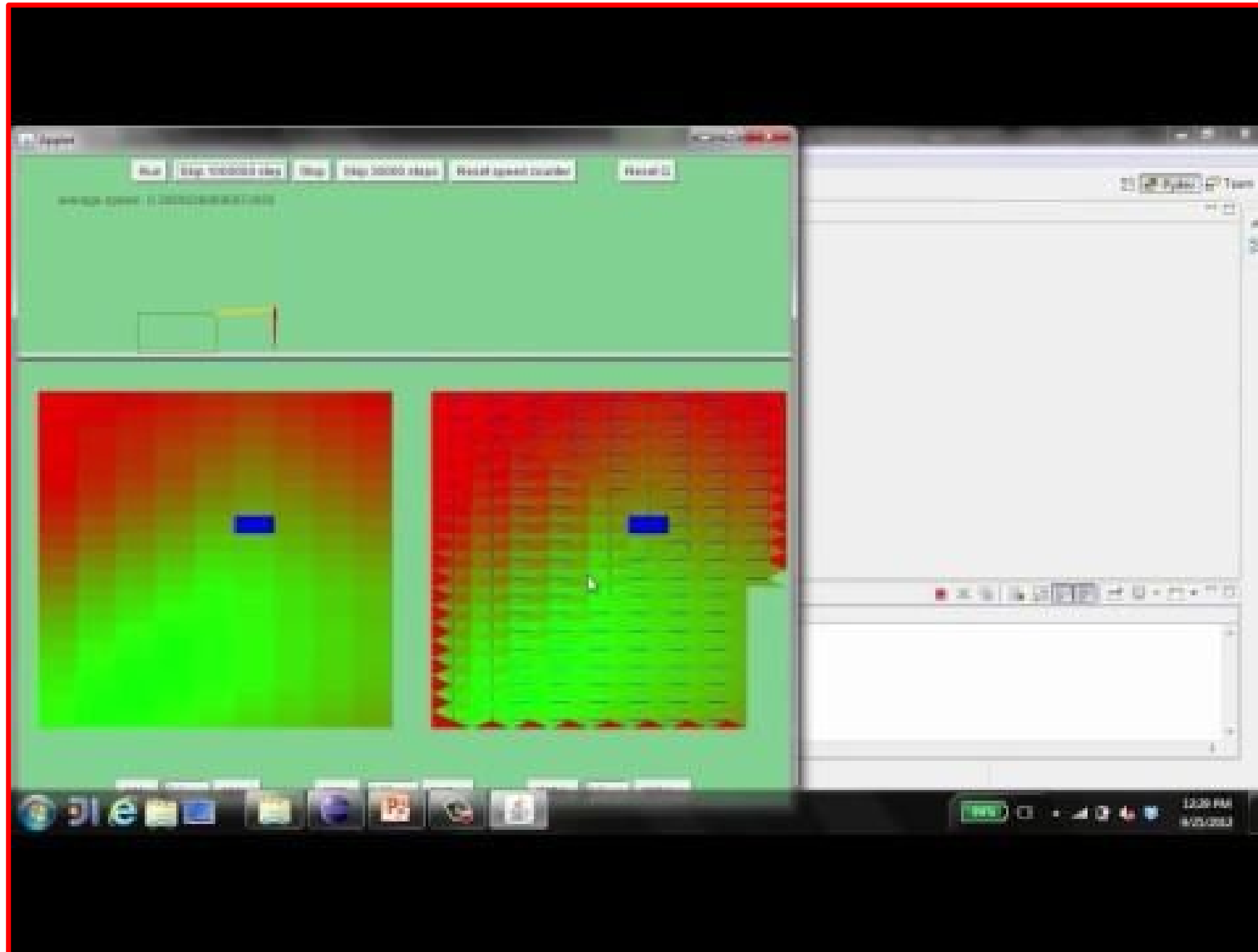


[Demo: Q-learning – gridworld (L10D2)]
 [Demo: Q-learning – crawler (L10D3)]

Video of Demo Q-Learning -- Gridworld

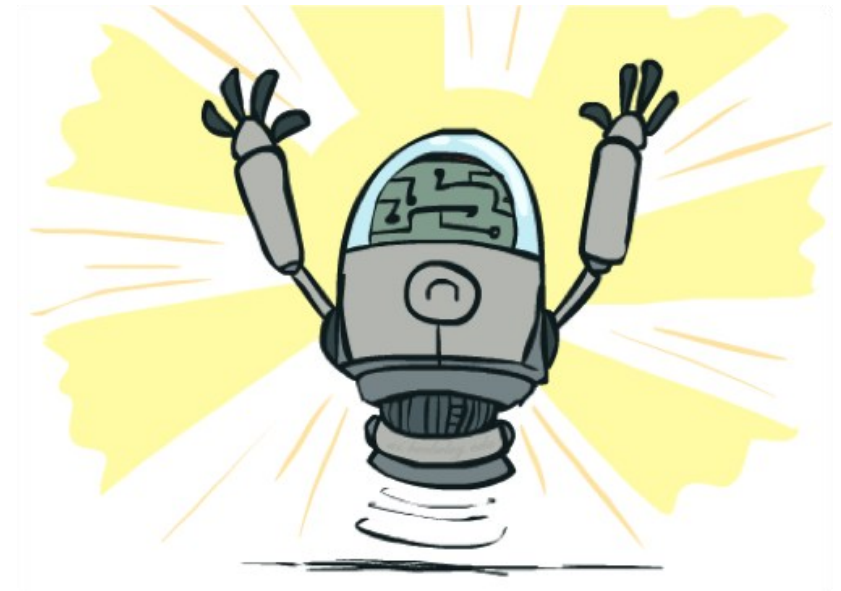


Video of Demo Q-Learning -- Crawler



Q-Learning Properties

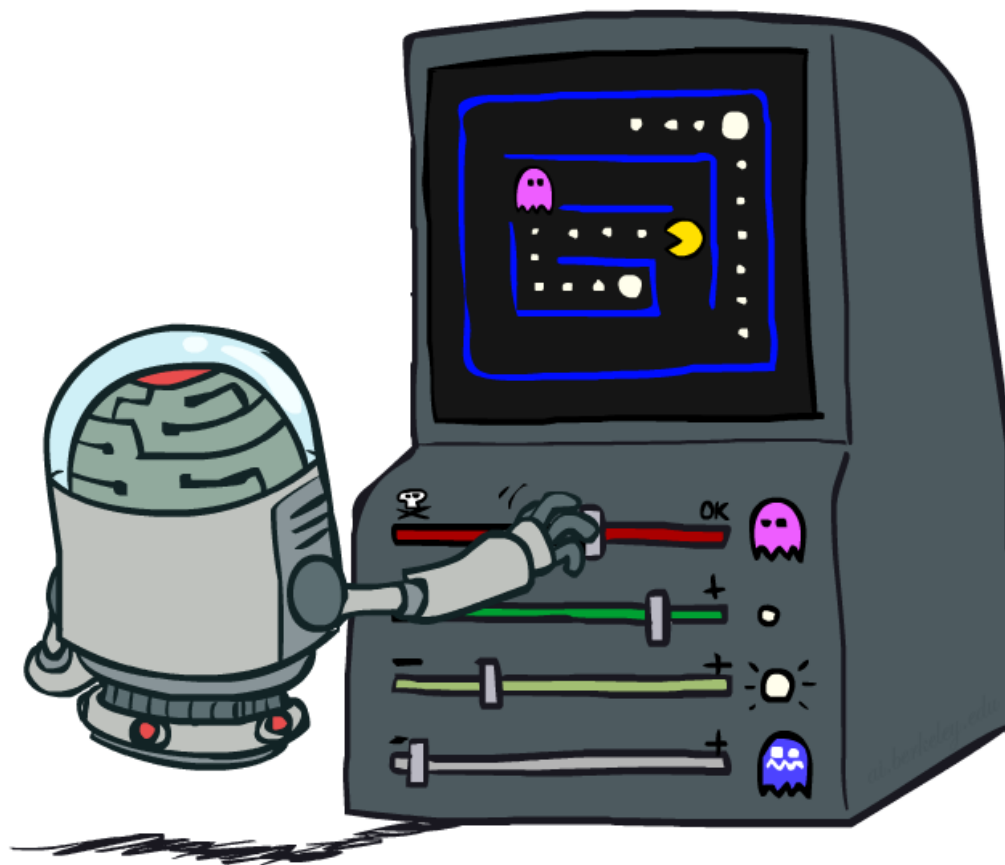
- Amazing result: Q-learning converges to optimal policy -- even if you're acting suboptimally!
- This is called **off-policy learning**
- Caveats:
 - You have to explore enough
 - You have to eventually make the learning rate small enough
 - ... but not decrease it too quickly
 - Basically, in the limit, it doesn't matter how you select actions (!)





CS 188: Artificial Intelligence

Reinforcement Learning II



Instructors: Dan Klein and Pieter Abbeel --- University of California, Berkeley

[These slides were created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. All CS188 materials are available at <http://ai.berkeley.edu>.]

Reinforcement Learning

- We still assume an MDP:
 - A set of states $s \in S$
 - A set of actions (per state) A
 - A model $T(s,a,s')$
 - A reward function $R(s,a,s')$
- Still looking for a policy $\pi(s)$
- New twist: don't know T or R , so must try out actions
- Big idea: Compute all averages over T using sample outcomes



The Story So Far: MDPs and RL

Known MDP: Offline Solution

Goal

Compute V^* , Q^* , π^*

Evaluate a fixed policy π

Technique

Value / policy iteration

Policy evaluation

Unknown MDP: Model-Based

Goal

Technique

Compute V^* , Q^* , π^*

VI/PI on approx. MDP

Evaluate a fixed policy π

PE on approx. MDP

Unknown MDP: Model-Free

Goal

Technique

Compute V^* , Q^* , π^*

Q-learning

Evaluate a fixed policy π

Value Learning

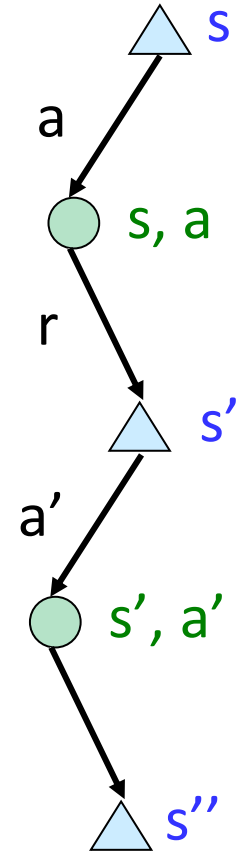
Model-Free Learning

- Model-free (temporal difference) learning

- Experience world through episodes

$$(s, a, r, s', a', r', s'', a'', r'', s'''' \dots)$$

- Update estimates each transition (s, a, r, s')
- Over time, updates will mimic Bellman updates



Q-Learning

- We'd like to do Q-value updates to each Q-state:

$$Q_{k+1}(s, a) \leftarrow \sum_{s'} T(s, a, s') \left[R(s, a, s') + \gamma \max_{a'} Q_k(s', a') \right]$$

- But can't compute this update without knowing T, R

- Instead, compute average as we go

- Receive a sample transition (s,a,r,s')
- This sample suggests

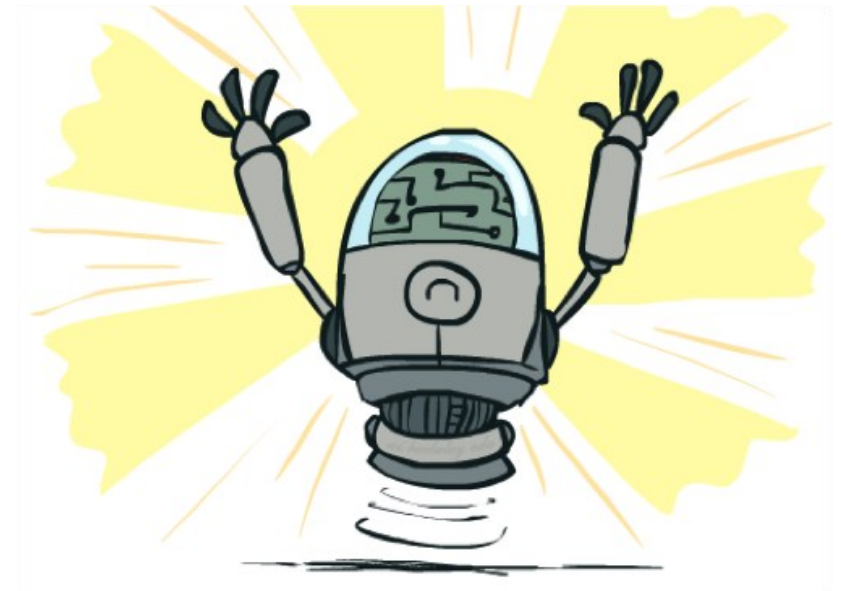
$$Q(s, a) \approx r + \gamma \max_{a'} Q(s', a')$$

- But we want to average over results from (s,a) (Why?)
- So keep a running average

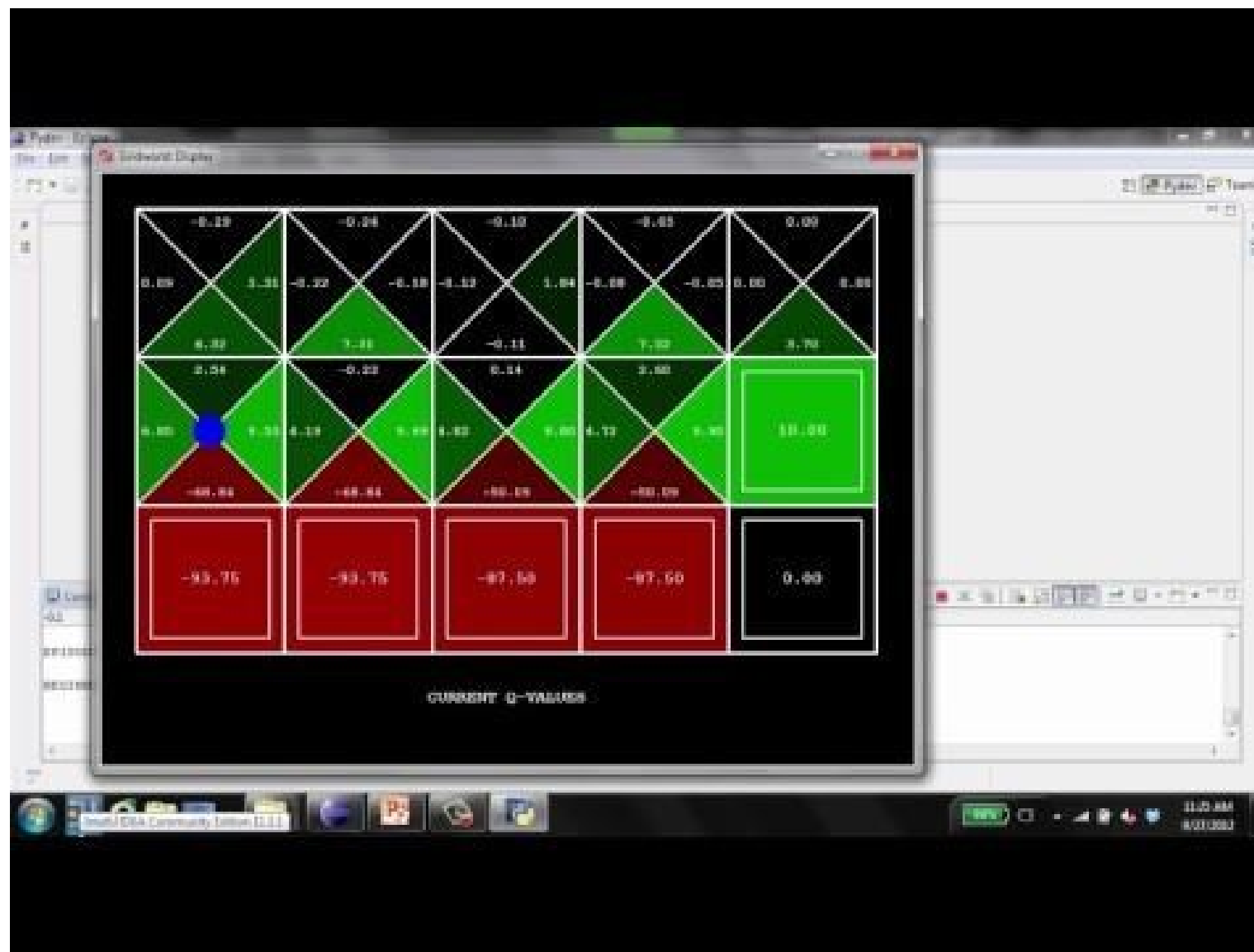
$$Q(s, a) \leftarrow (1 - \alpha)Q(s, a) + (\alpha) \left[r + \gamma \max_{a'} Q(s', a') \right]$$

Q-Learning Properties

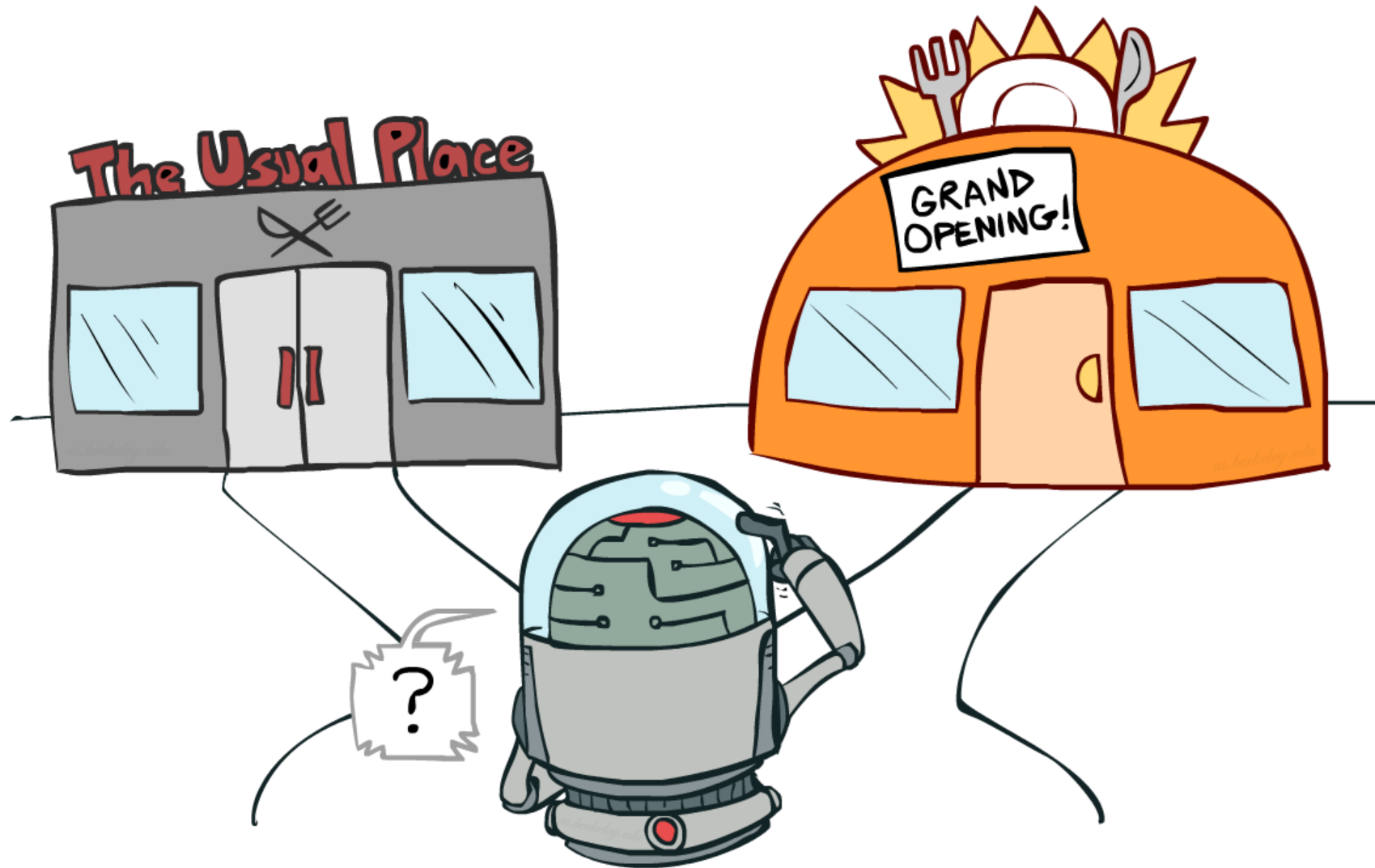
- Amazing result: Q-learning converges to optimal policy -- even if you're acting suboptimally!
- This is called **off-policy learning**
- Caveats:
 - You have to explore enough
 - You have to eventually make the learning rate small enough
 - ... but not decrease it too quickly
 - Basically, in the limit, it doesn't matter how you select actions (!)



Video of Demo Q-Learning Auto Cliff Grid



Exploration vs. Exploitation



How to Explore?

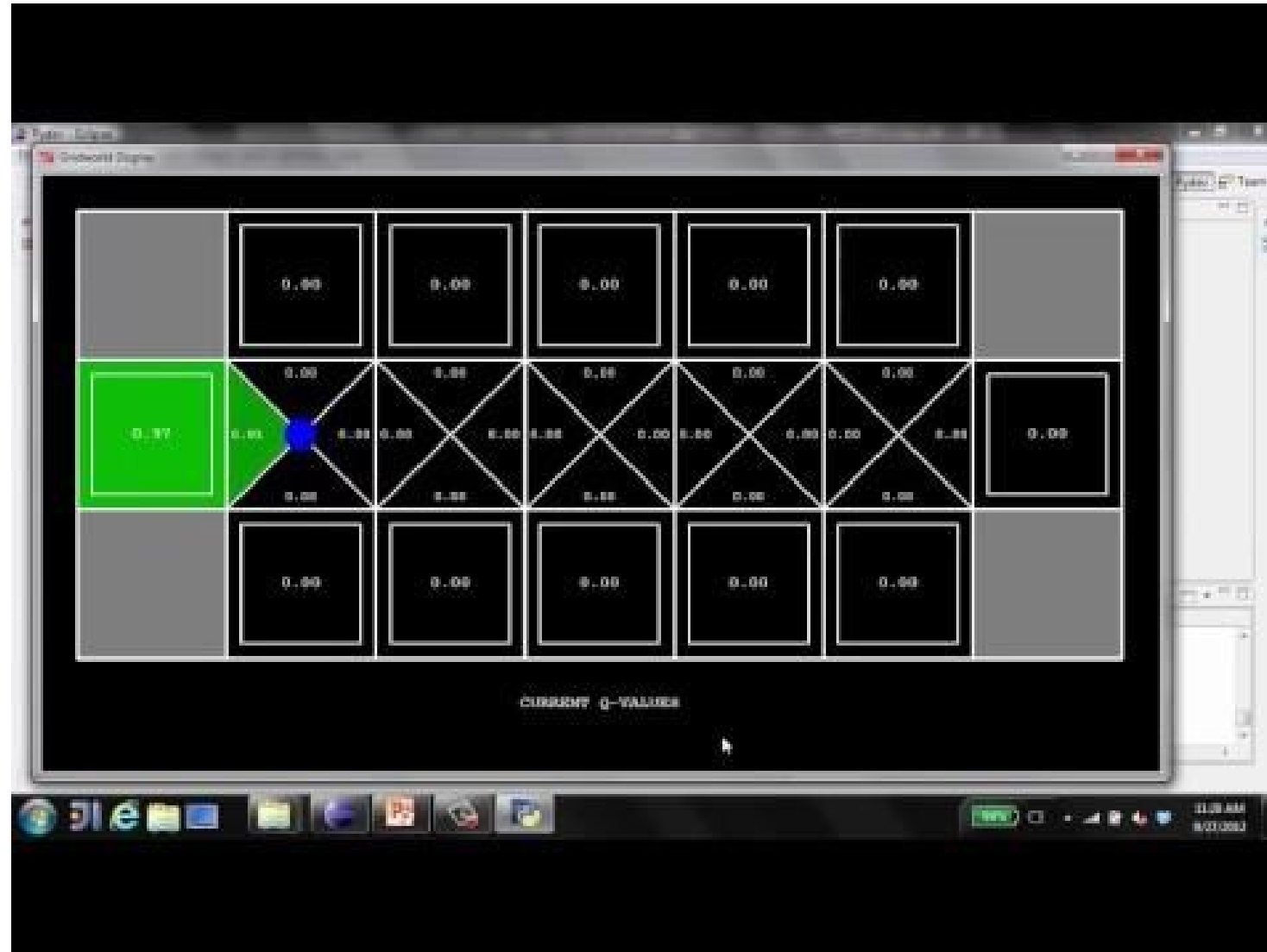
- Several schemes for forcing exploration
 - Simplest: random actions (ϵ -greedy)
 - Every time step, flip a coin
 - With (small) probability ϵ , act randomly
 - With (large) probability $1-\epsilon$, act on current policy
 - Problems with random actions?
 - You do eventually explore the space, but keep thrashing around once learning is done
 - One solution: lower ϵ over time
 - Another solution: exploration functions



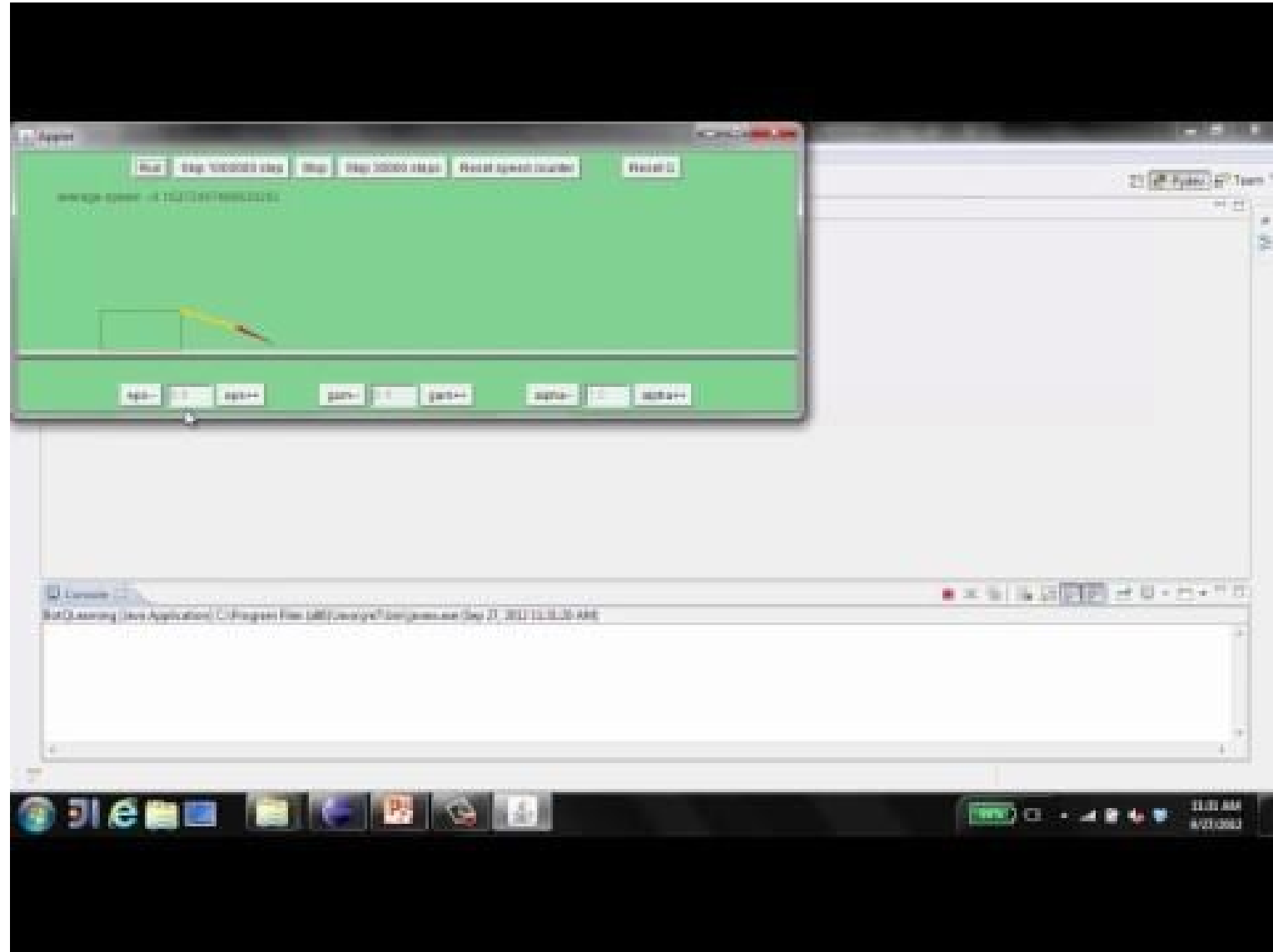
[Demo: Q-learning – manual exploration – bridge grid (L11D2)]

[Demo: Q-learning – epsilon-greedy -- crawler (L11D3)]

Video of Demo Q-learning – Manual Exploration – Bridge Grid



Video of Demo Q-learning – Epsilon-Greedy – Crawler



Exploration Functions

- When to explore?

- Random actions: explore a fixed amount
- Better idea: explore areas whose badness is not (yet) established, eventually stop exploring

- Exploration function

- Takes a value estimate u and a visit count n , and returns an optimistic utility, e.g. $f(u, n) = u + k/n$

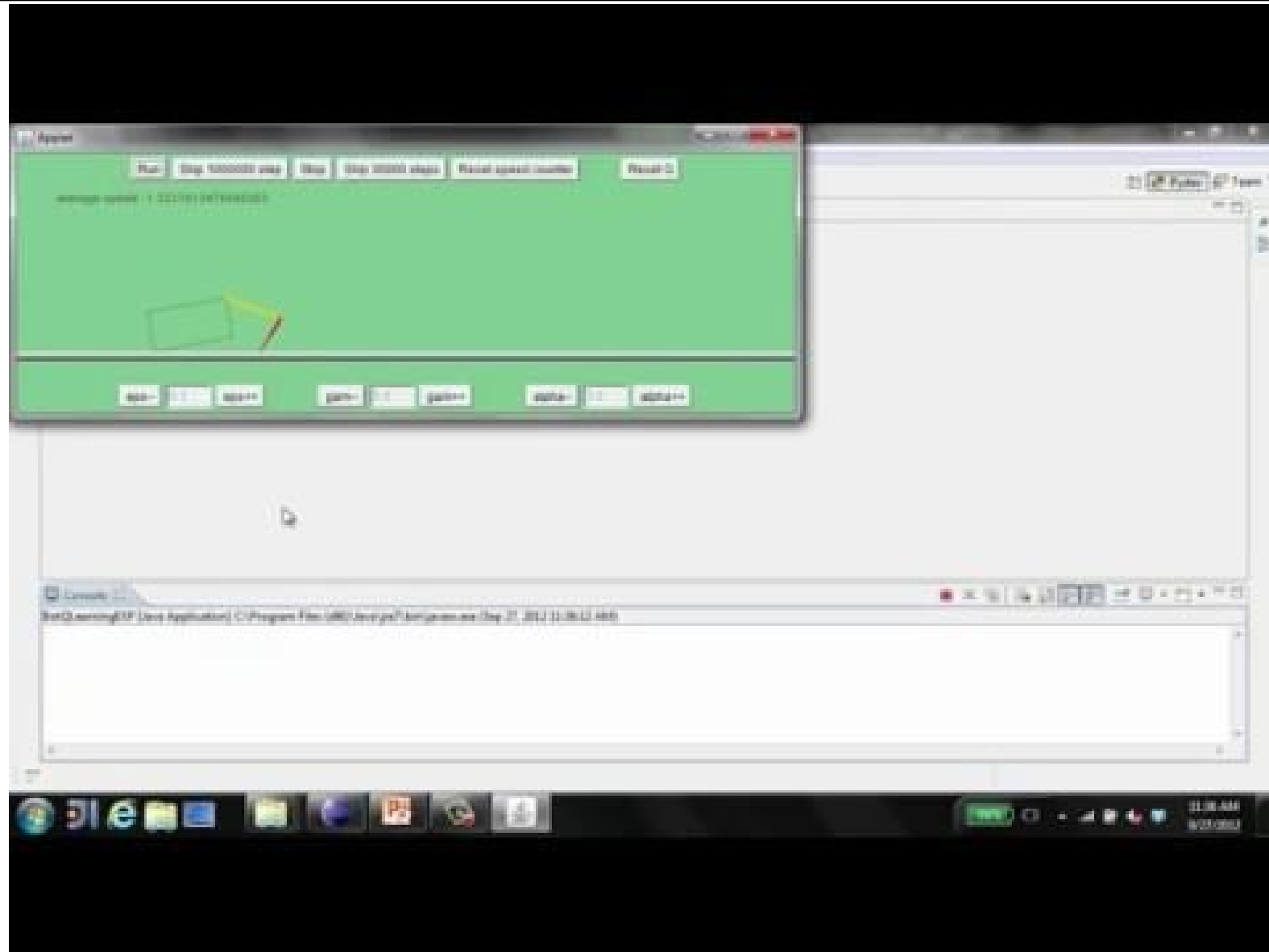
Regular Q-Update: $Q(s, a) \leftarrow_{\alpha} R(s, a, s') + \gamma \max_{a'} Q(s', a')$

Modified Q-Update: $Q(s, a) \leftarrow_{\alpha} R(s, a, s') + \gamma \max_{a'} f(Q(s', a'), N(s', a'))$

- Note: this propagates the “bonus” back to states that lead to unknown states as well!

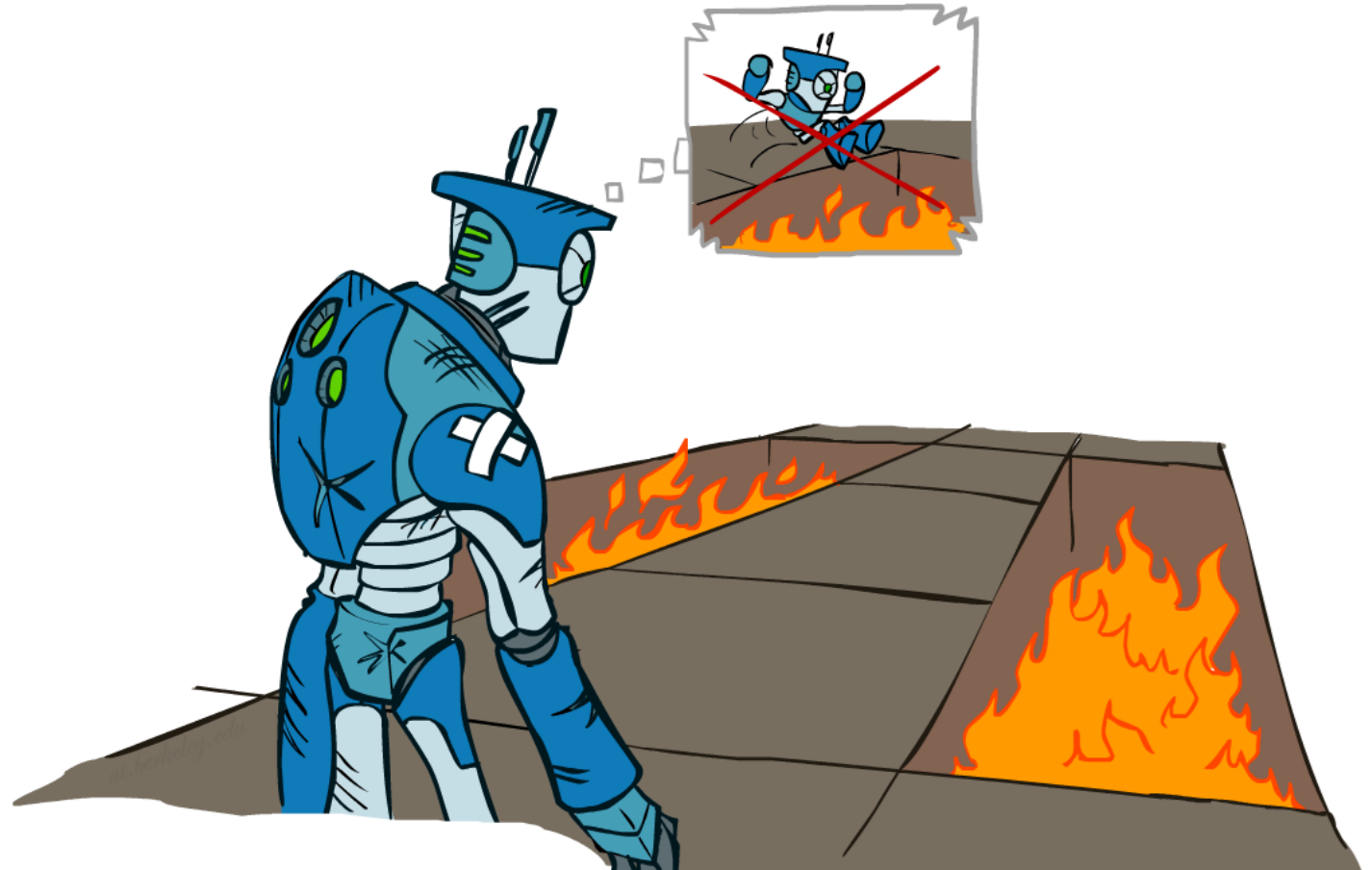


Video of Demo Q-learning – Exploration Function – Crawler

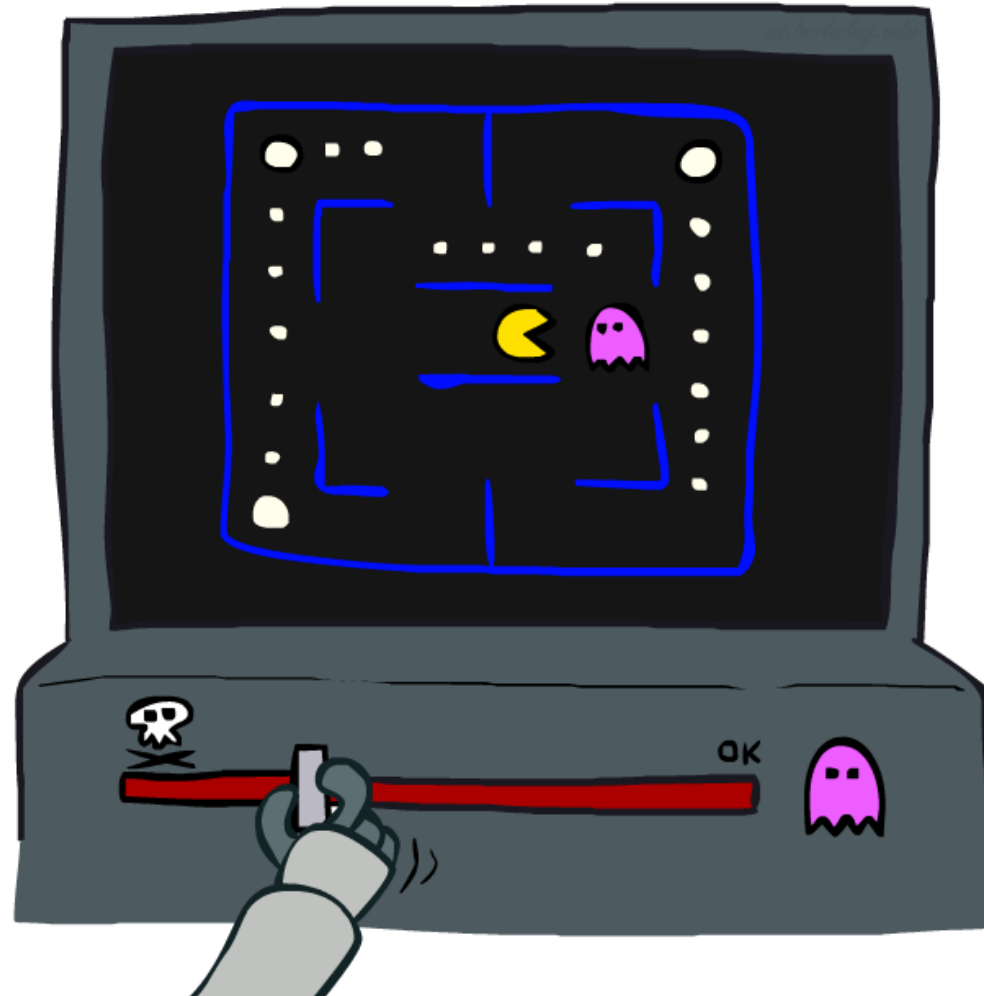


Regret

- Even if you learn the optimal policy, you still make mistakes along the way
- Regret is a measure of your total mistake cost: the difference between your (expected) rewards, including youthful suboptimality, and optimal (expected) rewards
- Minimizing regret goes beyond learning to be optimal – it requires optimally learning to be optimal
- Example: random exploration and exploration functions both end up optimal, but random exploration has higher regret

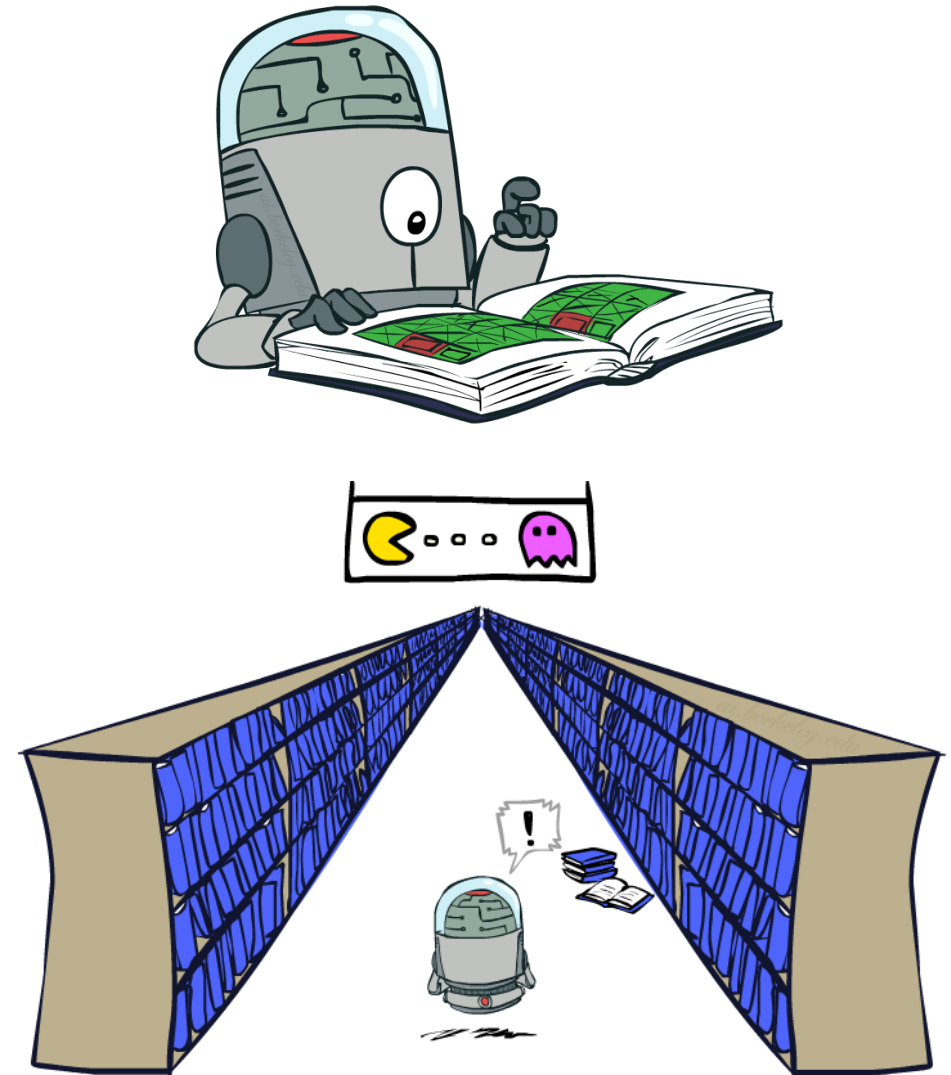


Approximate Q-Learning



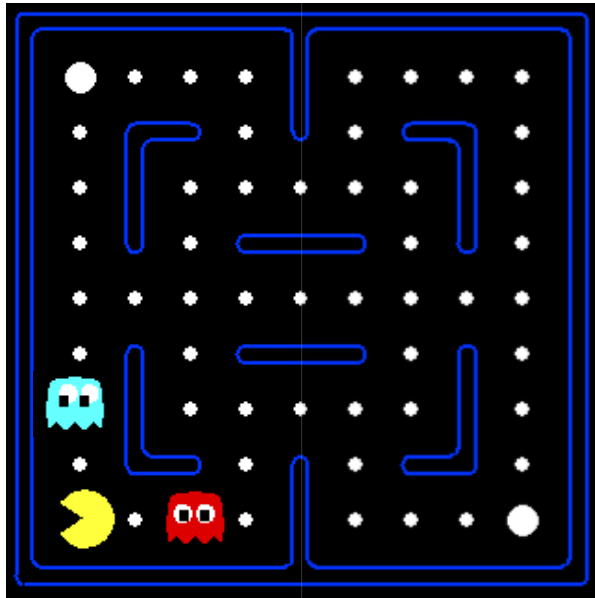
Generalizing Across States

- Basic Q-Learning keeps a table of all q-values
- In realistic situations, we cannot possibly learn about every single state!
 - Too many states to visit them all in training
 - Too many states to hold the q-tables in memory
- Instead, we want to generalize:
 - Learn about some small number of training states from experience
 - Generalize that experience to new, similar situations
 - This is a fundamental idea in machine learning, and we'll see it over and over again

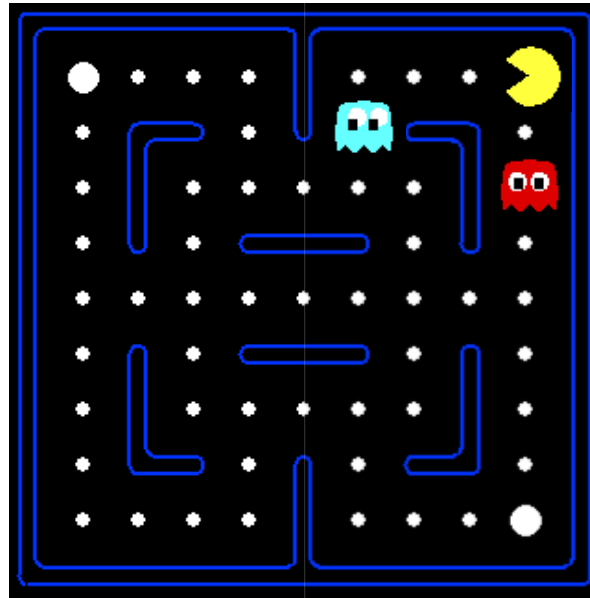


Example: Pacman

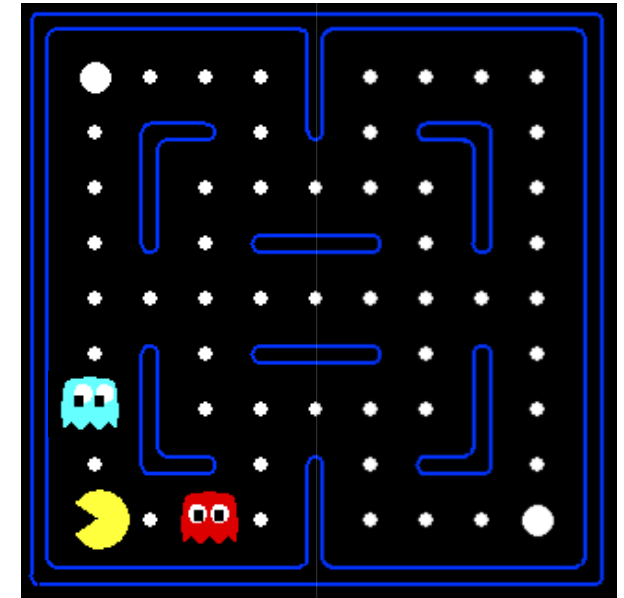
Let's say we discover through experience that this state is bad:



In naïve q-learning, we know nothing about this state:

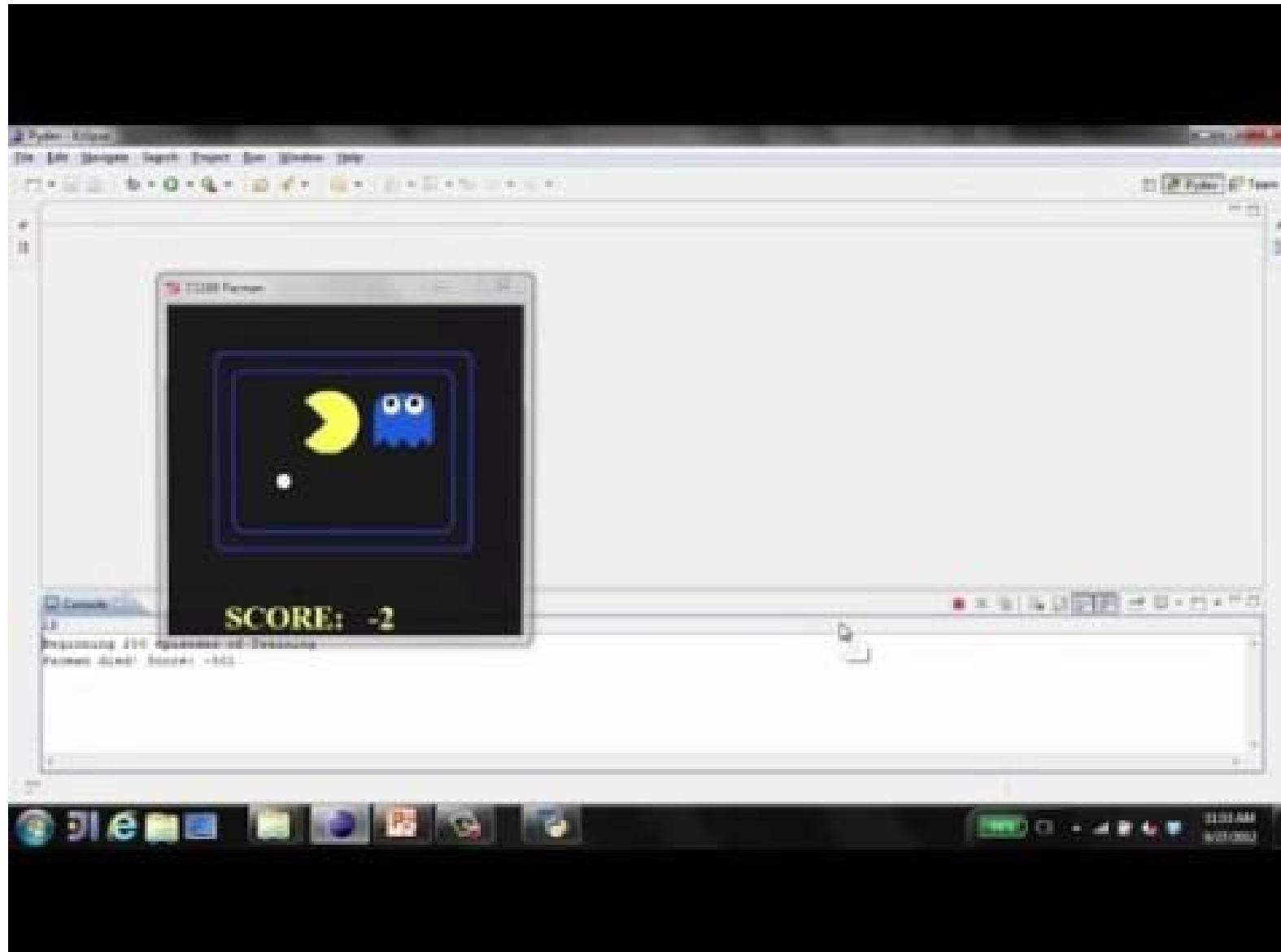


Or even this one!

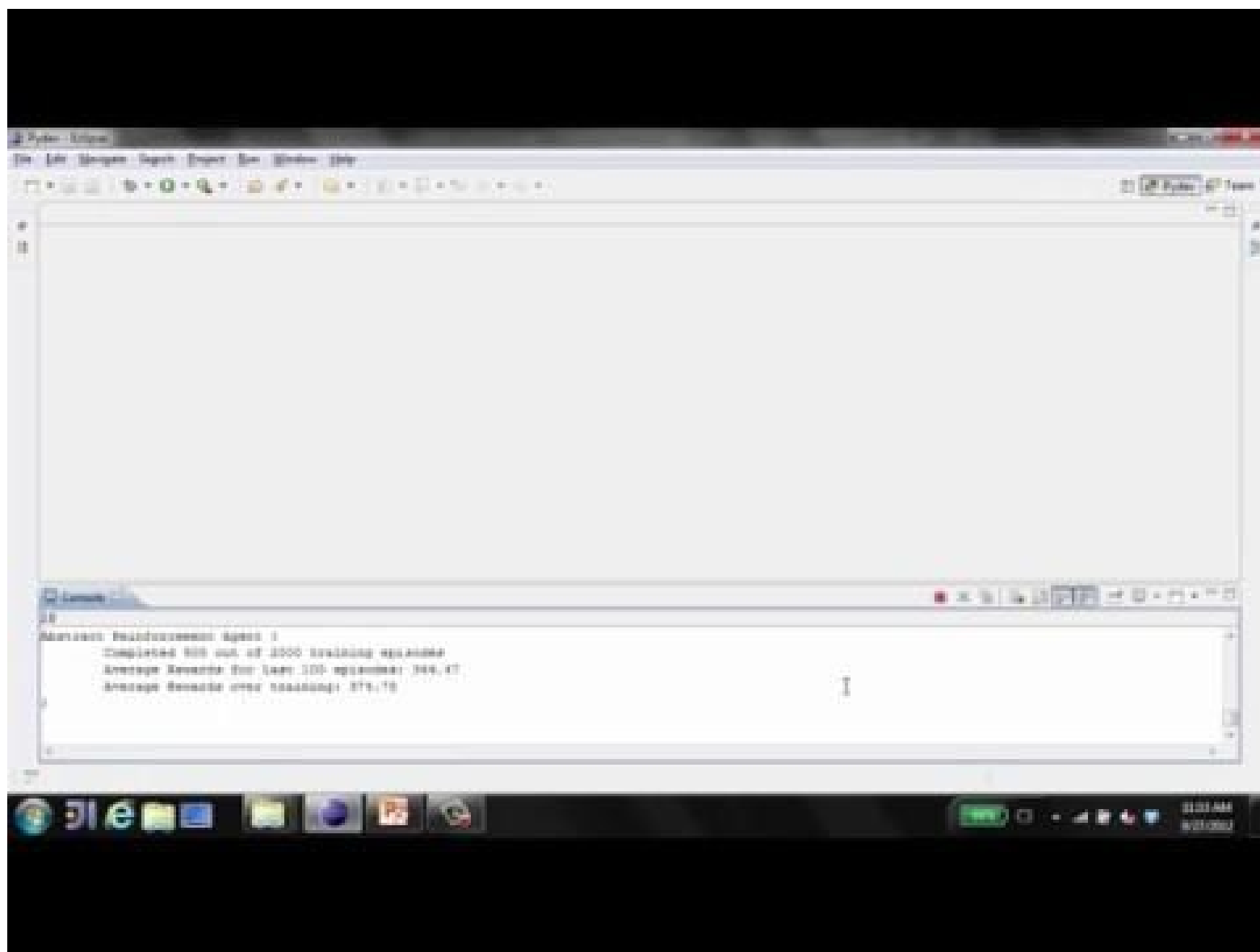


[Demo: Q-learning – pacman – tiny – watch all (L11D5)]
[Demo: Q-learning – pacman – tiny – silent train (L11D6)]
[Demo: Q-learning – pacman – tricky – watch all (L11D7)]

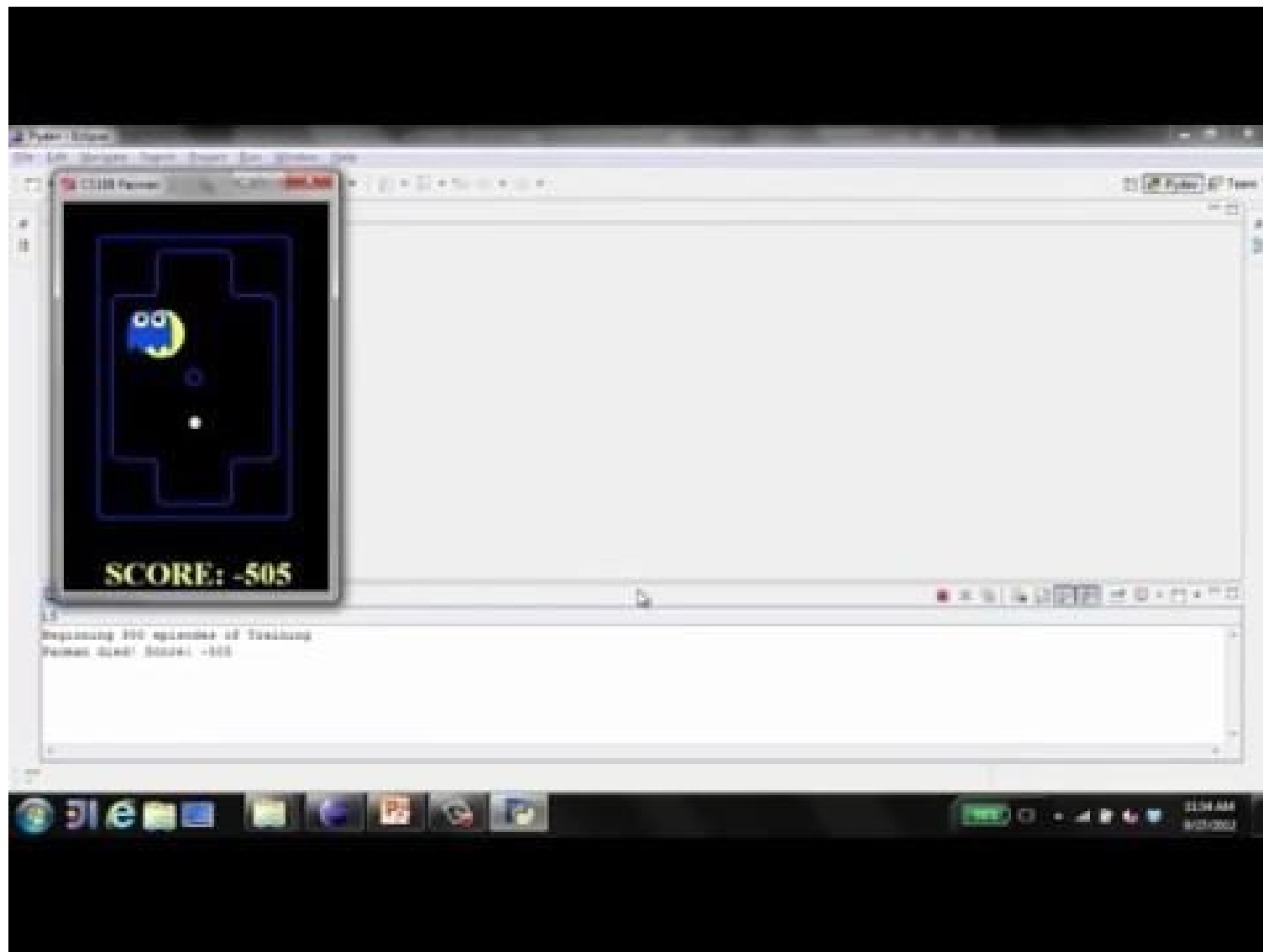
Video of Demo Q-Learning Pacman – Tiny – Watch All



Video of Demo Q-Learning Pacman – Tiny – Silent Train

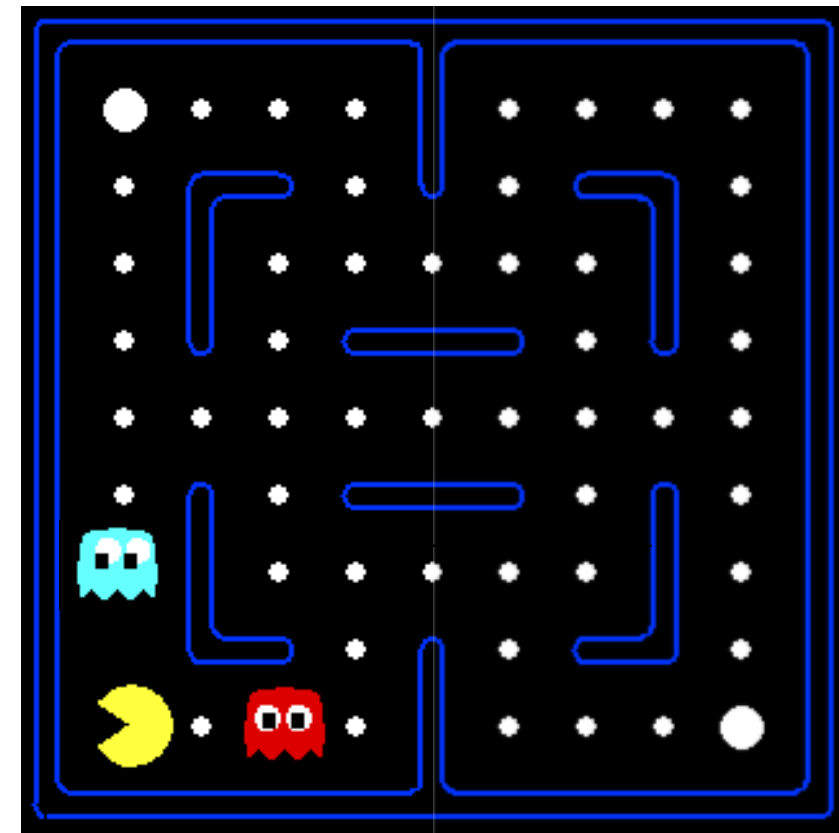


Video of Demo Q-Learning Pacman – Tricky – Watch All



Feature-Based Representations

- Solution: describe a state using a vector of features (properties)
 - Features are functions from states to real numbers (often 0/1) that capture important properties of the state
 - Example features:
 - Distance to closest ghost
 - Distance to closest dot
 - Number of ghosts
 - $1 / (\text{dist to dot})^2$
 - Is Pacman in a tunnel? (0/1)
 - etc.
 - Is it the exact state on this slide?
 - Can also describe a q-state (s, a) with features (e.g. action moves closer to food)



Linear Value Functions

- Using a feature representation, we can write a q function (or value function) for any state using a few weights:

$$V(s) = w_1 f_1(s) + w_2 f_2(s) + \dots + w_n f_n(s)$$

$$Q(s, a) = w_1 f_1(s, a) + w_2 f_2(s, a) + \dots + w_n f_n(s, a)$$

- Advantage: our experience is summed up in a few powerful numbers
- Disadvantage: states may share features but actually be very different in value!

Approximate Q-Learning

$$Q(s, a) = w_1 f_1(s, a) + w_2 f_2(s, a) + \dots + w_n f_n(s, a)$$

- Q-learning with linear Q-functions:

$$\text{transition} = (s, a, r, s')$$

$$\text{difference} = \left[r + \gamma \max_{a'} Q(s', a') \right] - Q(s, a)$$

$$Q(s, a) \leftarrow Q(s, a) + \alpha [\text{difference}]$$

$$w_i \leftarrow w_i + \alpha [\text{difference}] f_i(s, a)$$

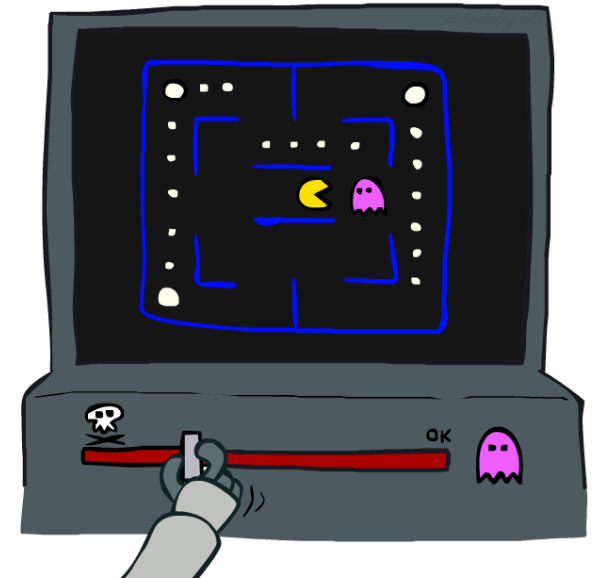
Exact Q's

Approximate Q's

- Intuitive interpretation:

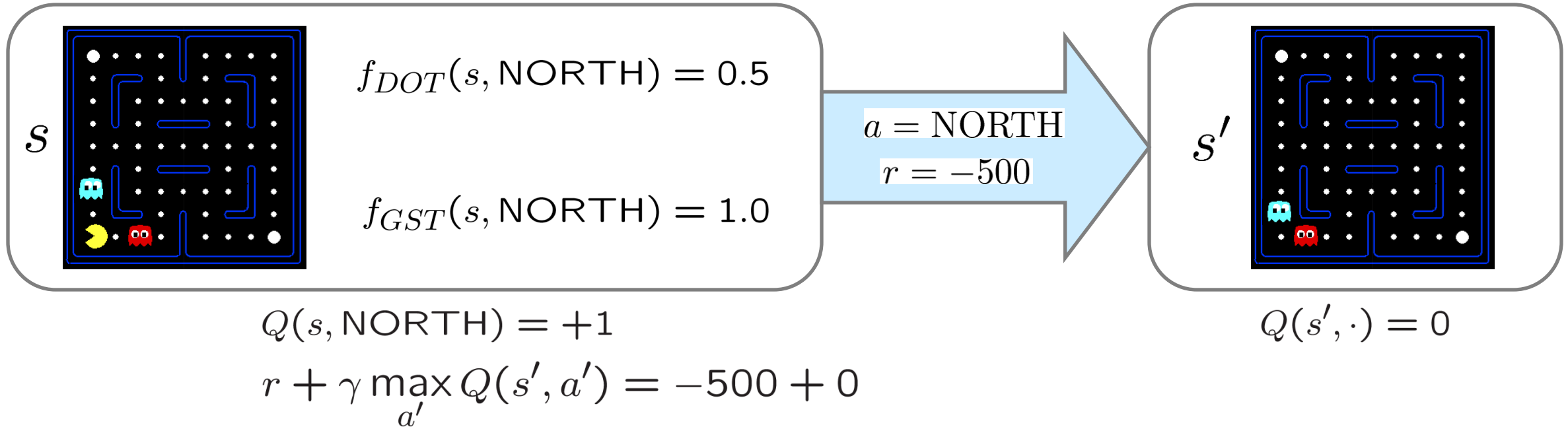
- Adjust weights of active features
- E.g., if something unexpectedly bad happens, blame the features that were on: disprefer all states with that state's features

- Formal justification: online least squares



Example: Q-Pacman

$$Q(s, a) = 4.0 f_{DOT}(s, a) - 1.0 f_{GST}(s, a)$$



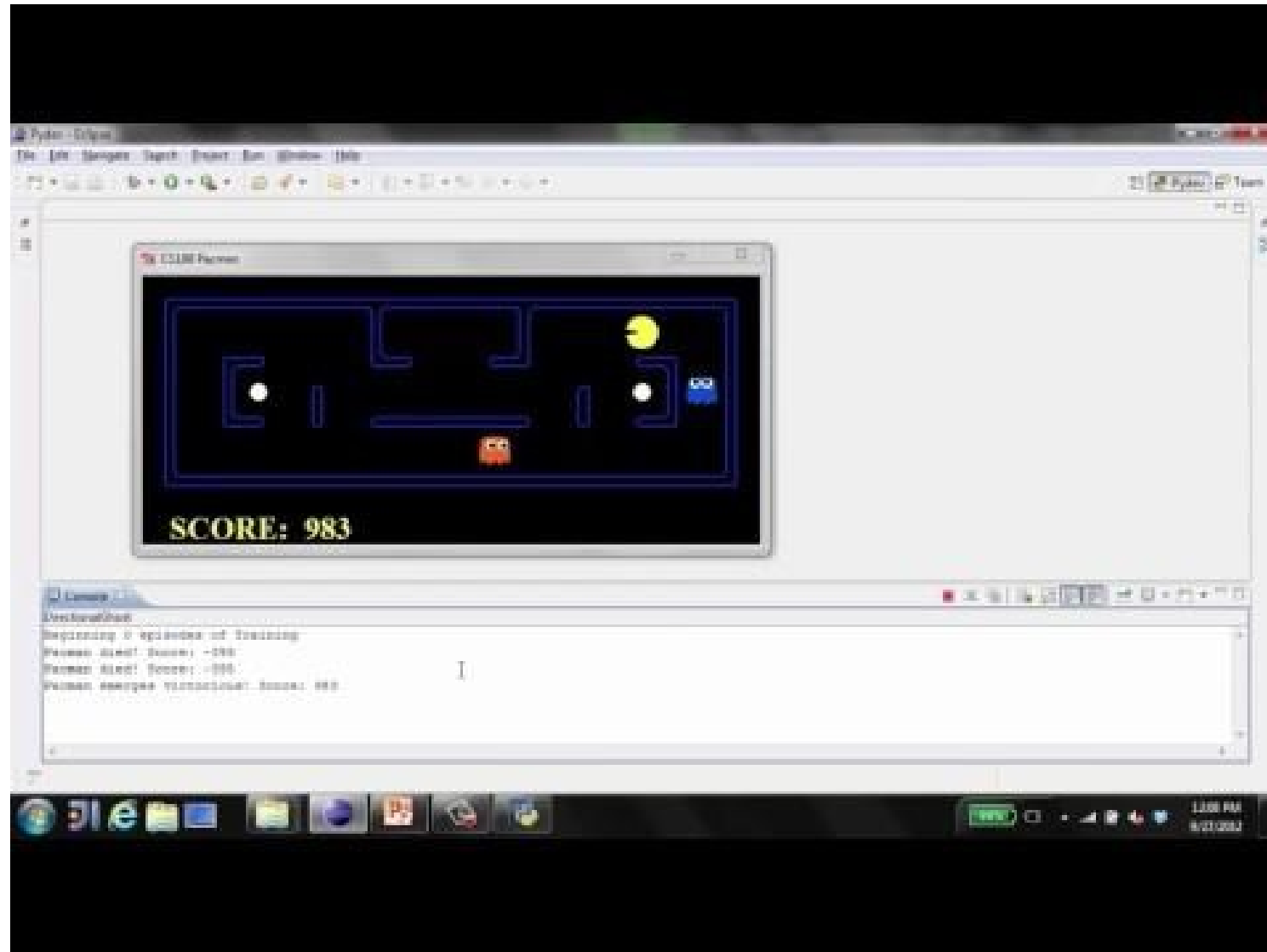
difference = -501 $\longrightarrow$

$$w_{DOT} \leftarrow 4.0 + \alpha [-501] 0.5$$

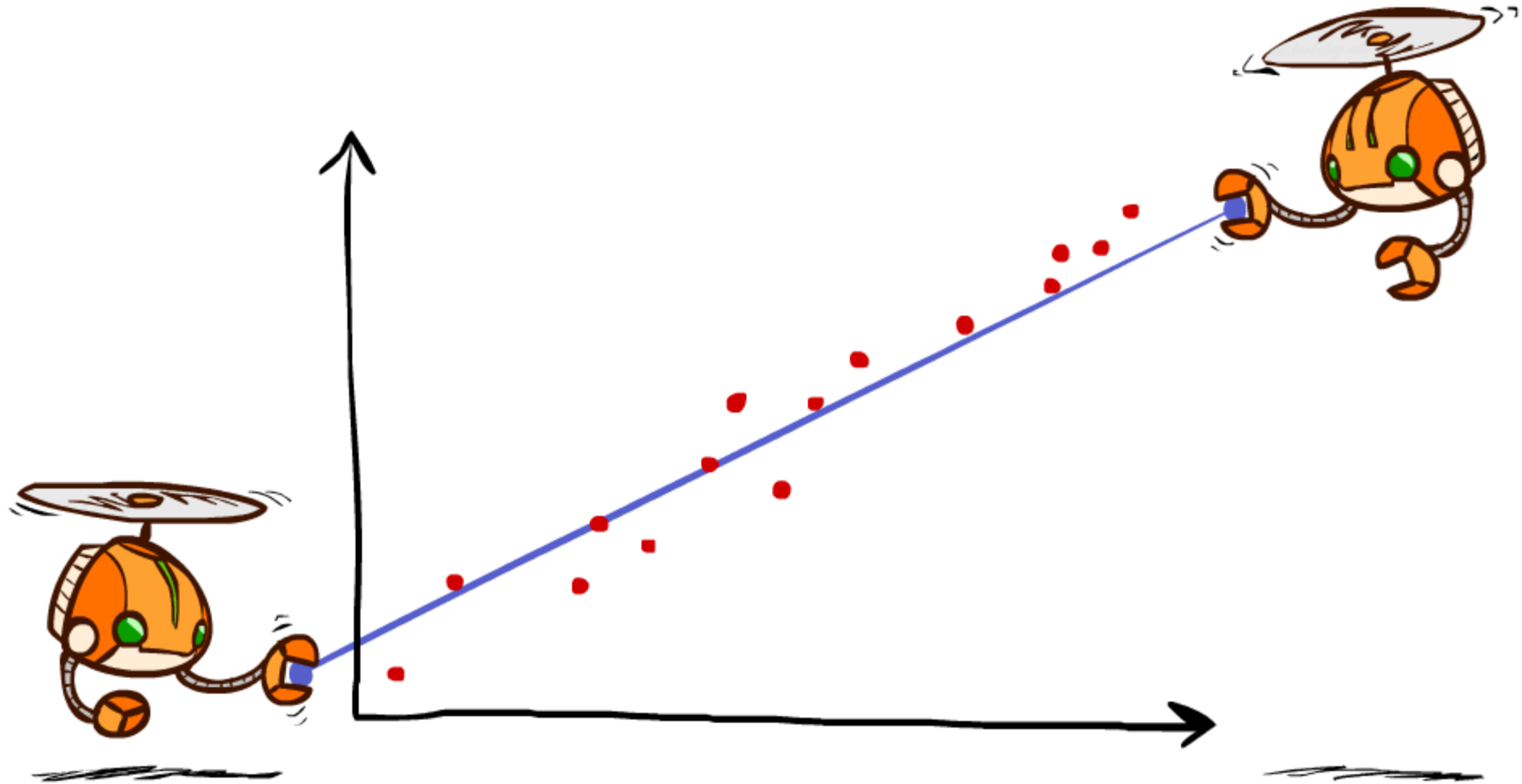
$$w_{GST} \leftarrow -1.0 + \alpha [-501] 1.0$$

$$Q(s, a) = 3.0 f_{DOT}(s, a) - 3.0 f_{GST}(s, a)$$

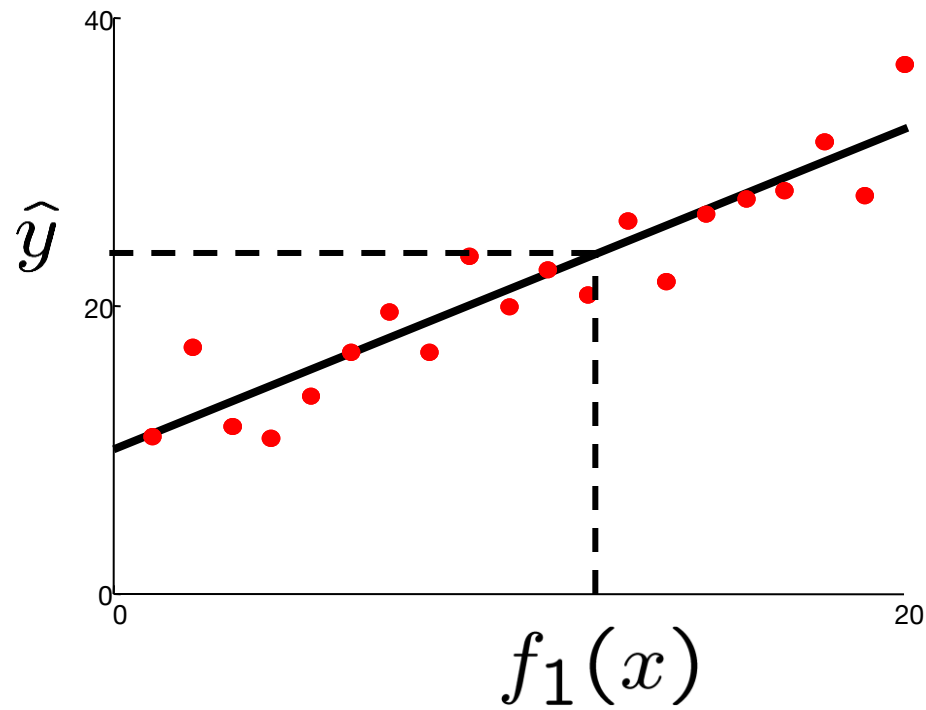
Video of Demo Approximate Q-Learning -- Pacman



Q-Learning and Least Squares

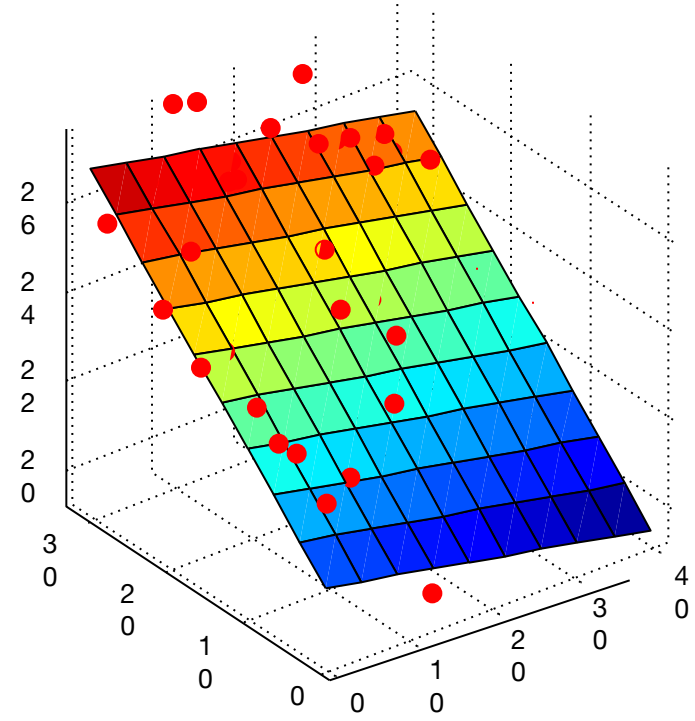


Linear Approximation: Regression*



Prediction:

$$\hat{y} = w_0 + w_1 f_1(x)$$

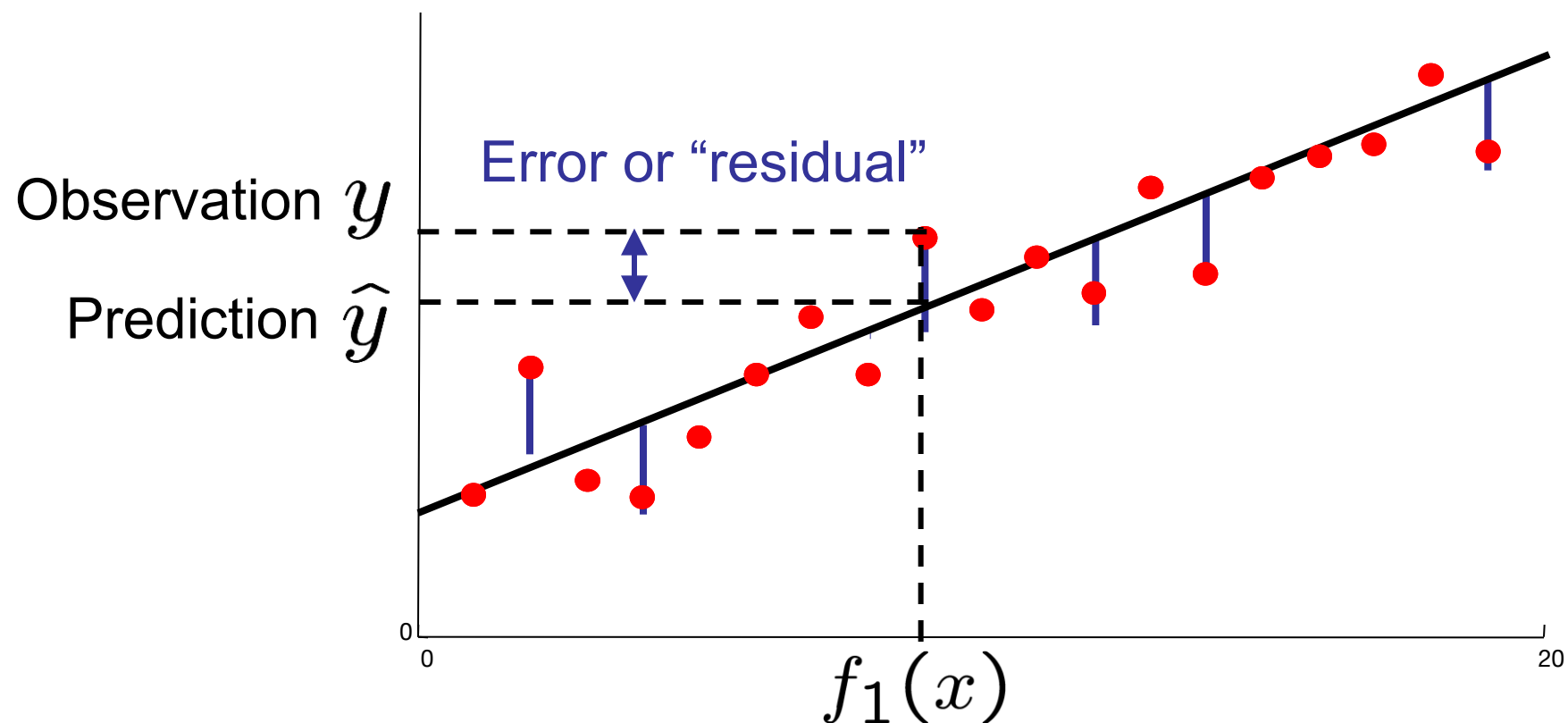


Prediction:

$$\hat{y}_i = w_0 + w_1 f_1(x) + w_2 f_2(x)$$

Optimization: Least Squares*

$$\text{total error} = \sum_i (y_i - \hat{y}_i)^2 = \sum_i \left(y_i - \sum_k w_k f_k(x_i) \right)^2$$



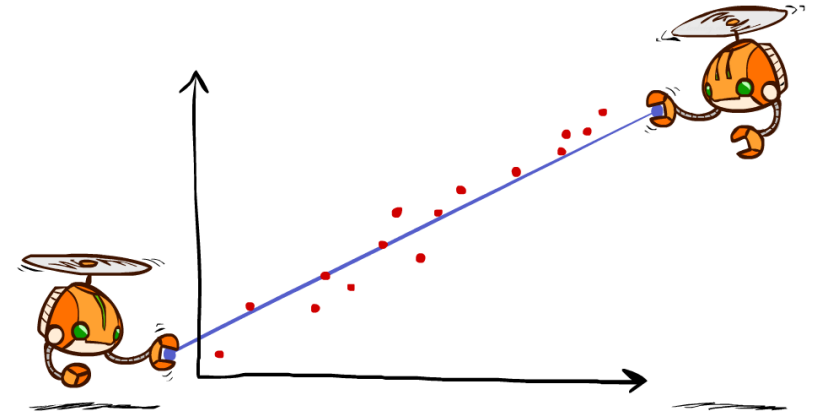
Minimizing Error*

Imagine we had only one point x , with features $f(x)$, target value y , and weights w :

$$\text{error}(w) = \frac{1}{2} \left(y - \sum_k w_k f_k(x) \right)^2$$

$$\frac{\partial \text{error}(w)}{\partial w_m} = - \left(y - \sum_k w_k f_k(x) \right) f_m(x)$$

$$w_m \leftarrow w_m + \alpha \left(y - \sum_k w_k f_k(x) \right) f_m(x)$$



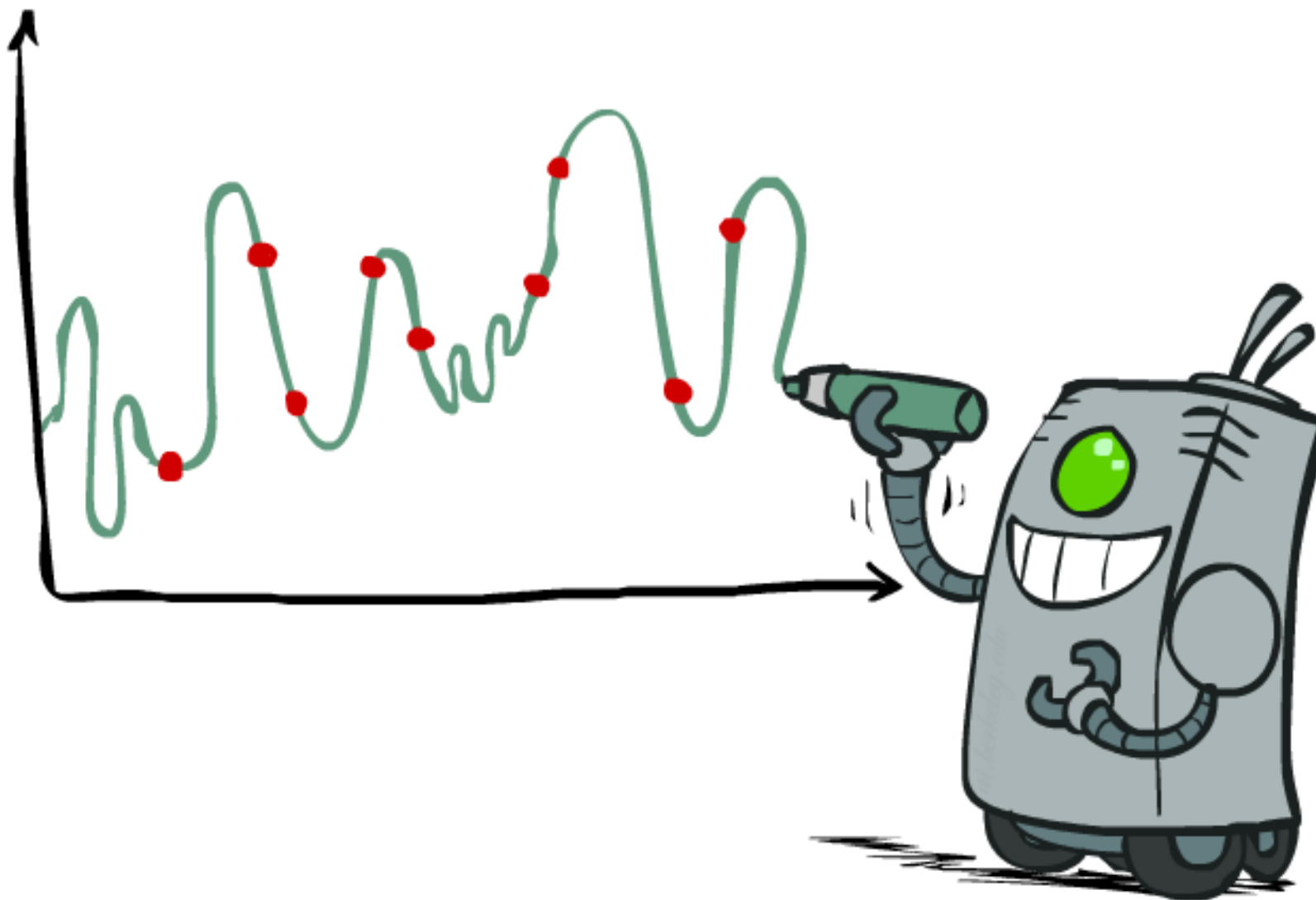
Approximate q update explained:

$$w_m \leftarrow w_m + \alpha \left[r + \gamma \max_a Q(s', a') - Q(s, a) \right] f_m(s, a)$$

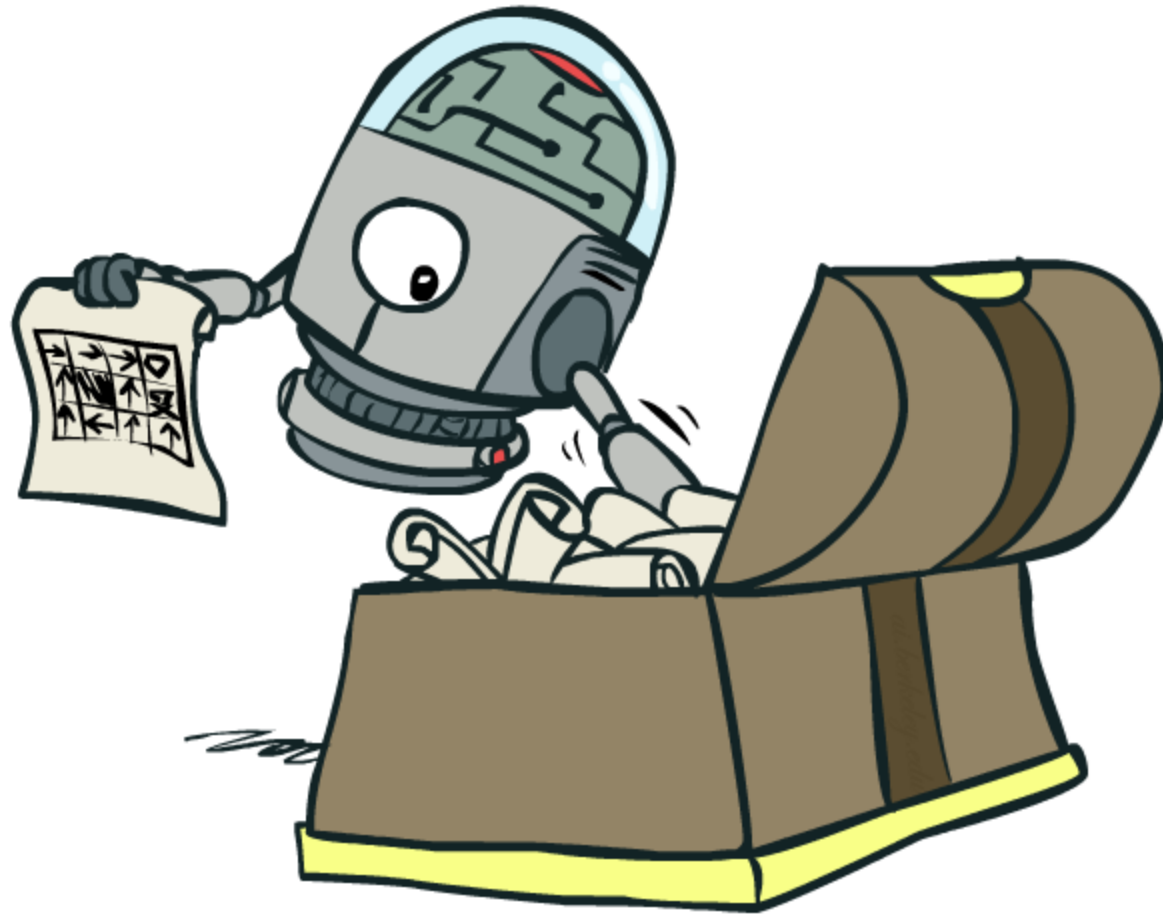
“target”

“prediction”

Overfitting: Why Limiting Capacity Can Help*



Policy Search



Policy Search

- Problem: often the feature-based policies that work well (win games, maximize utilities) aren't the ones that approximate V / Q best
 - E.g. your value functions from project 2 were probably horrible estimates of future rewards, but they still produced good decisions
 - Q-learning's priority: get Q-values close (modeling)
 - Action selection priority: get ordering of Q-values right (prediction)
 - We'll see this distinction between modeling and prediction again later in the course
- Solution: learn policies that maximize rewards, not the values that predict them
- Policy search: start with an ok solution (e.g. Q-learning) then fine-tune by hill climbing on feature weights

Policy Search

- Simplest policy search:
 - Start with an initial linear value function or Q-function
 - Nudge each feature weight up and down and see if your policy is better than before
- Problems:
 - How do we tell the policy got better?
 - Need to run many sample episodes!
 - If there are a lot of features, this can be impractical
- Better methods exploit lookahead structure, sample wisely, change multiple parameters...

Policy Search

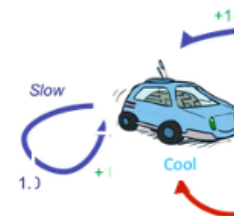


-

Whiteboard used in lectorial

MODEL-FREE
Learning:
Passive vs
Active
learning

- Still assume a Markov decision process (MDP):
 - A set of states $s \in S$
 - A set of actions (per state) A
 - ~~A model $T(s,a,s')$~~
 - ~~A reward function $R(s,a,s')$~~
- Still looking for a policy $\pi(s)$



Passive Learning
Fix the policy π
We learn $V^\pi(s)$
Direct Evolution
ADP
Variation of
VI

TD-learning

Active Learning

Learn policy π
work with
 $Q(s,a)$

TD-learning

TD on-learning

Exploration
vs
Exploitation

Questions?

Questions for discussion

- What are the main components of a Markov Decision Process (MDP), and how do they relate to reinforcement learning?
- What is the primary objective of an agent in reinforcement learning, and how is it achieved?
- What is the difference between model-free and model-based reinforcement learning algorithms?
- How does the concept of exploration vs. exploitation affect the learning process in reinforcement learning?